



GE Medical Systems

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Technical Publication

Direction 5113145-100

Revision 03

GE Medical Systems LightSpeed 5.X Functional Interconnects

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER’S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER’S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N’EST DISPONIBLE QU’EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L’ANGLAIS, C’EST AU CLIENT QU’IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D’INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N’A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L’OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。
GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “damage in shipment” written on all copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm>
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.



CAUTION
Risk of
Explosion



ATTENTION
Danger
d’Explosion

LITHIUM BATTERY CAUTIONARY STATEMENTS

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer’s instructions.

Il y a danger d’explosion s’il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d’un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS PQR Process to report all omissions, errors, and defects in this publication.

Revision History

Rev	Date	Reason for change
1	06/07/04	Initial Release
2	04/21/05	PQR 13034160; PQR 13034162
3	07/21/05	PQR 13036875

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Chapter 1

System Communications

Section 1.0 — Slip Ring Communications Interface - 2399621

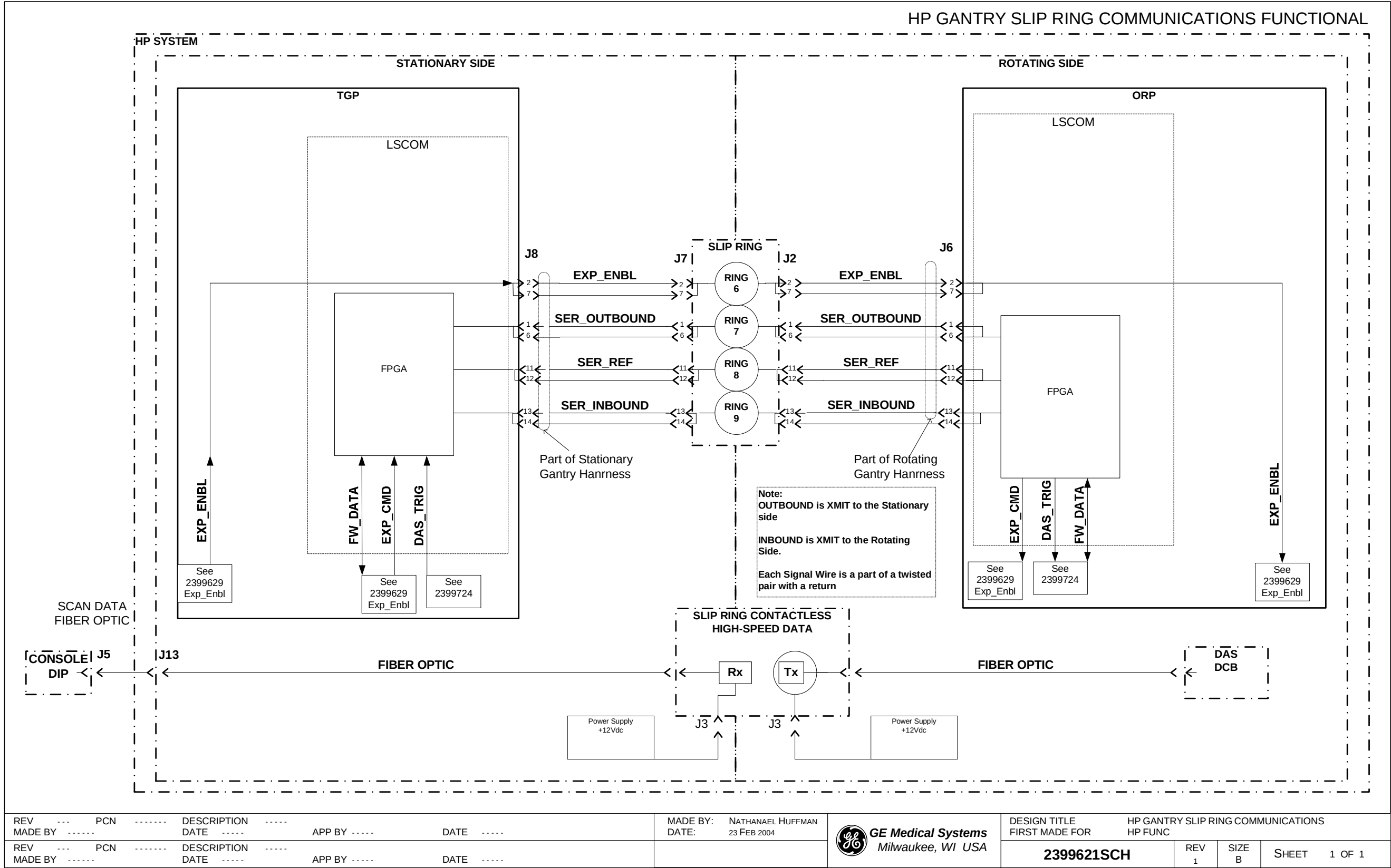


Figure 1-1 2399621 Sheet 1, Rev. 1

Section 2.0 — Final Scan Control, Console Ethernet - 2399622

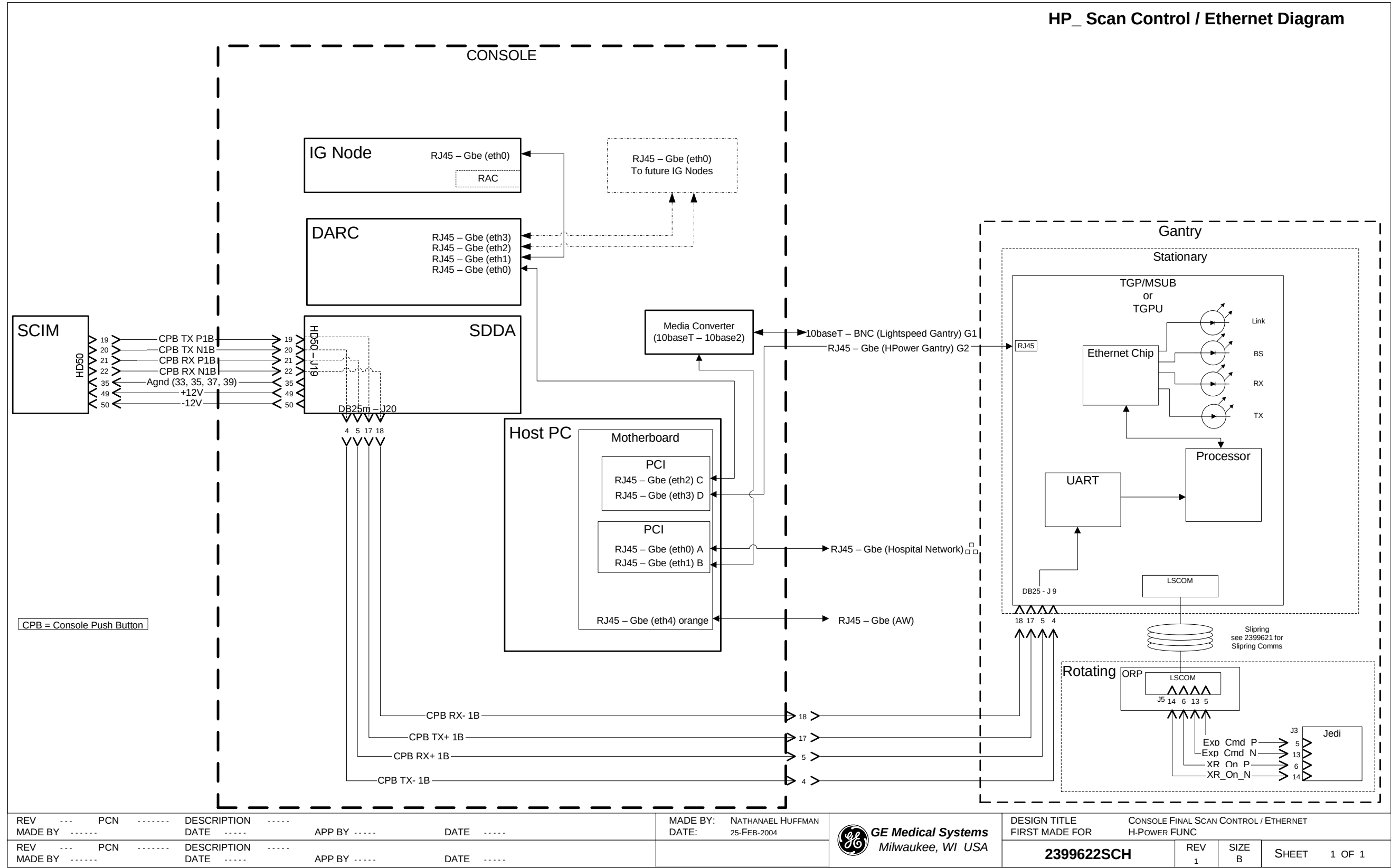


Figure 1-2 2399622 Sheet 1, Rev 1

Section 3.0 — Rotating Controller Interface Bus (RCIB) - 2399623

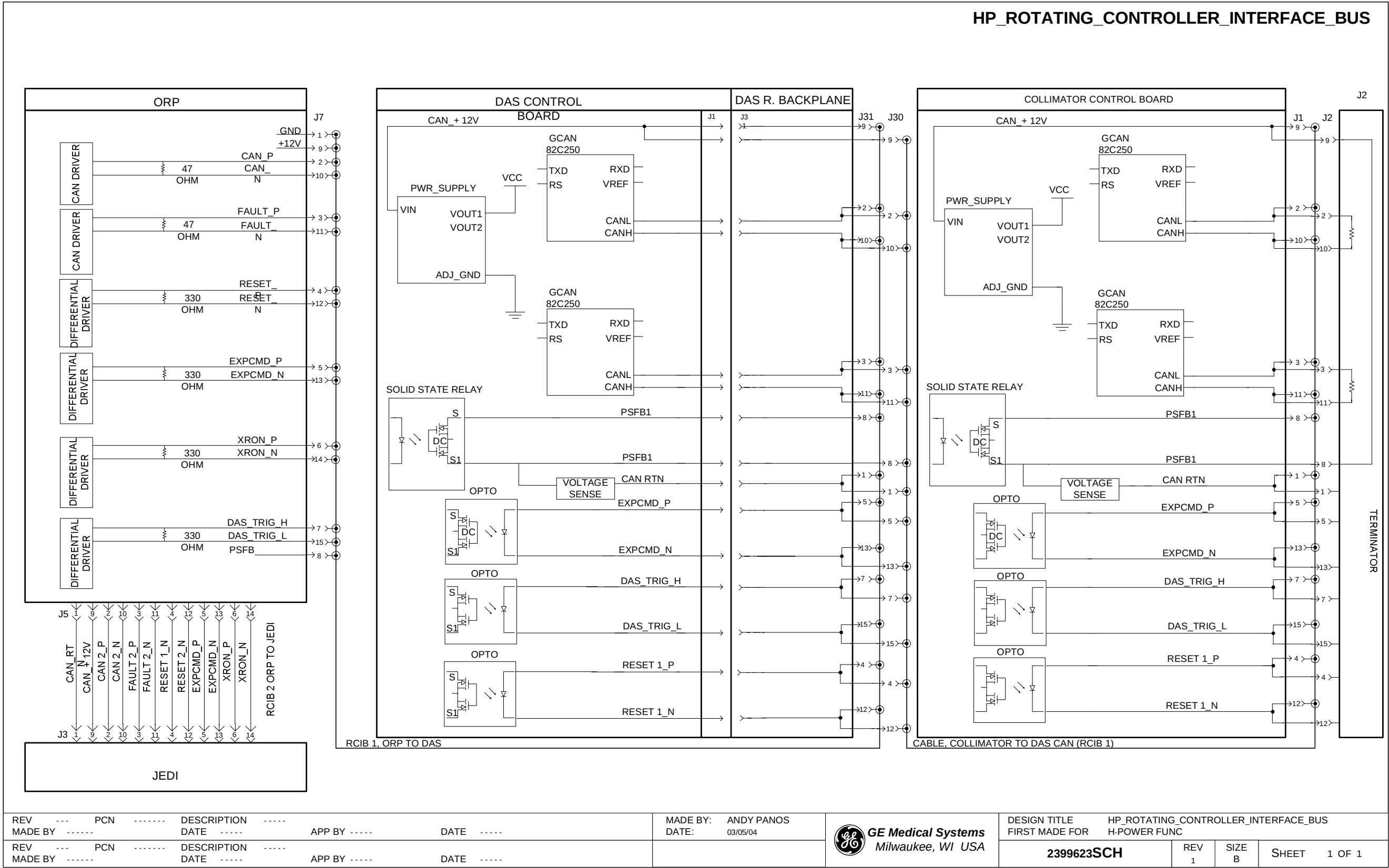


Figure 1-3 2399623 Sheet 1, Rev. 1

Chapter 2

X-Ray Generation

Section 2.0 — HP60 JEDI Tube Fans/Pump/Smart ID Interface - 5141517

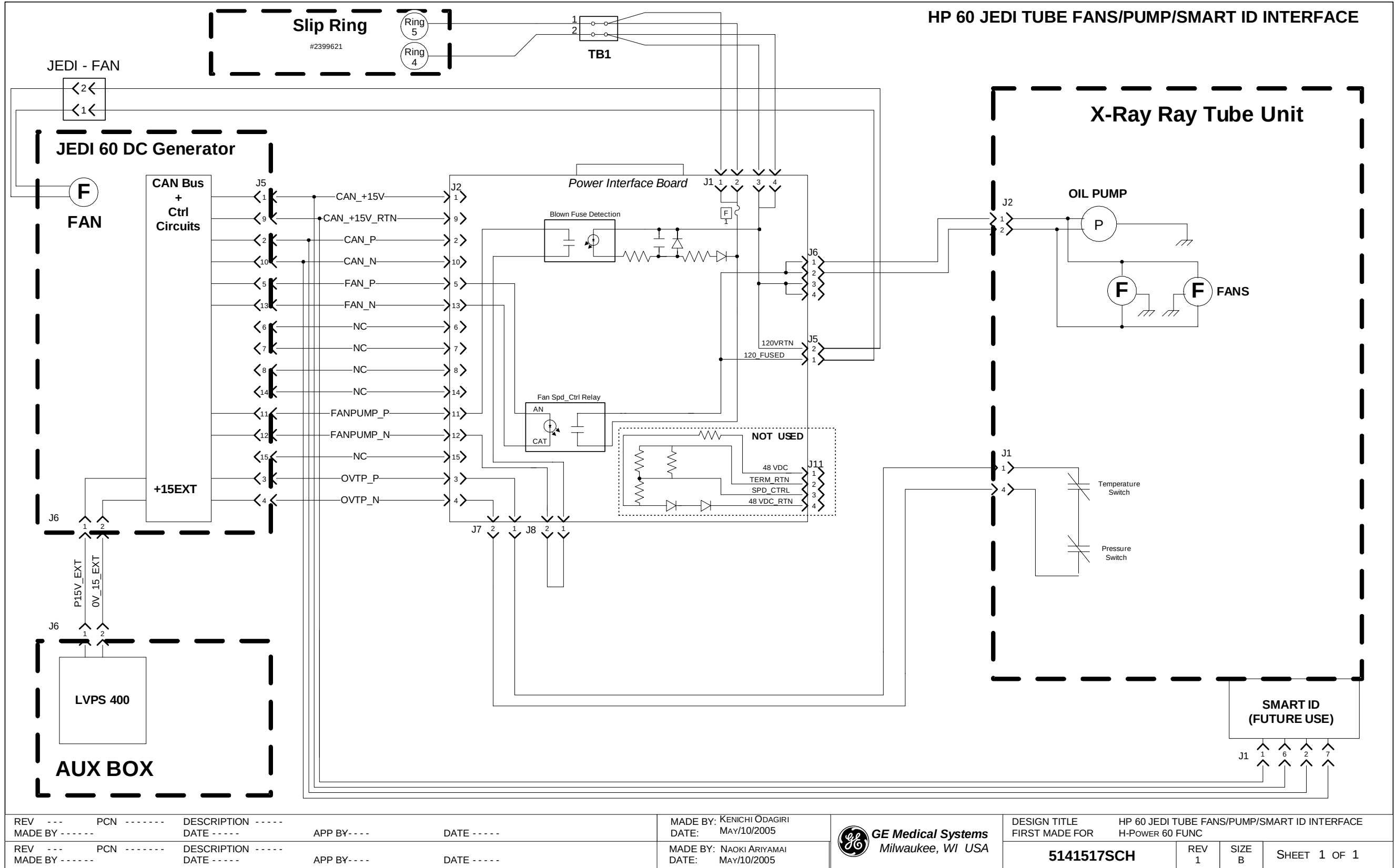


Figure 2-2 5141517 Sheet 1, Rev 1

Section 3.0 — JEDI Interface - kV Loop - 2399625

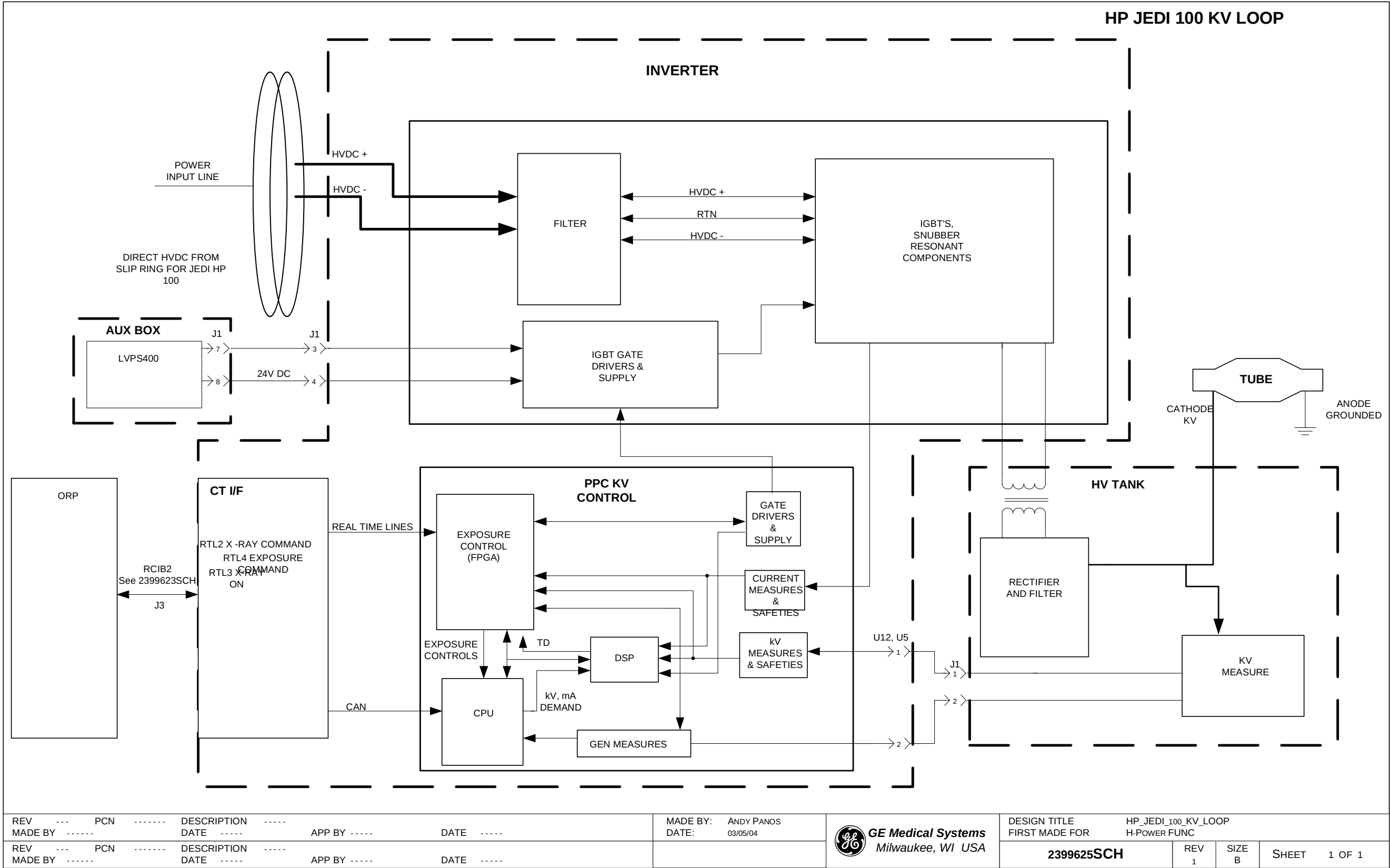


Figure 2-3 2399625 Sheet 1, Rev. 1

Section 4.0 — HP60 JEDI kV Loop - 5141524

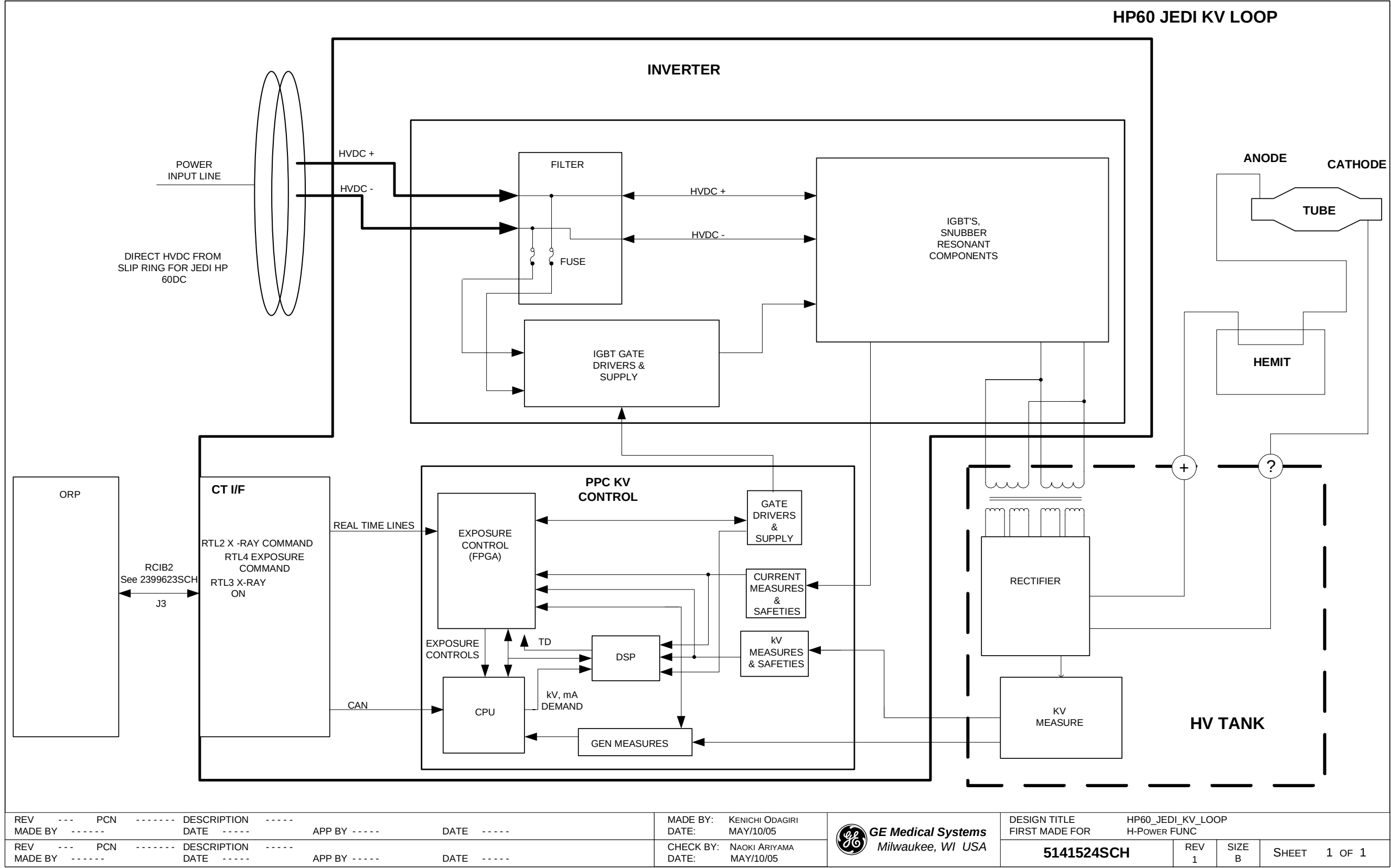


Figure 2-4 5141524 Sheet 1, Rev 1

Section 5.0 — JEDI Interface - mA Loop - 2399628

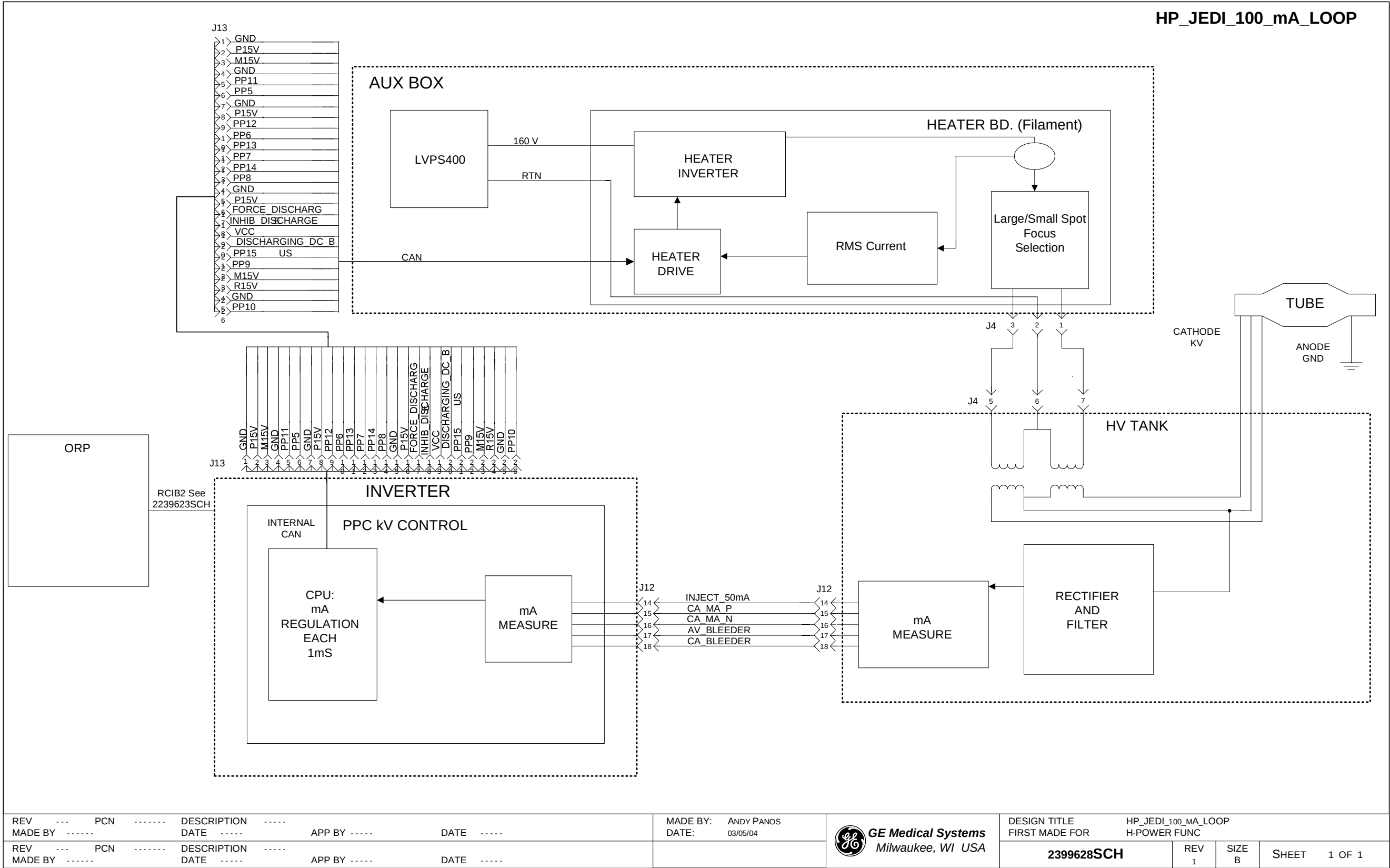


Figure 2-5 2399628 Sheet 1, Rev. 1

Section 6.0 — HP60 JEDI mA Loop - 5141527

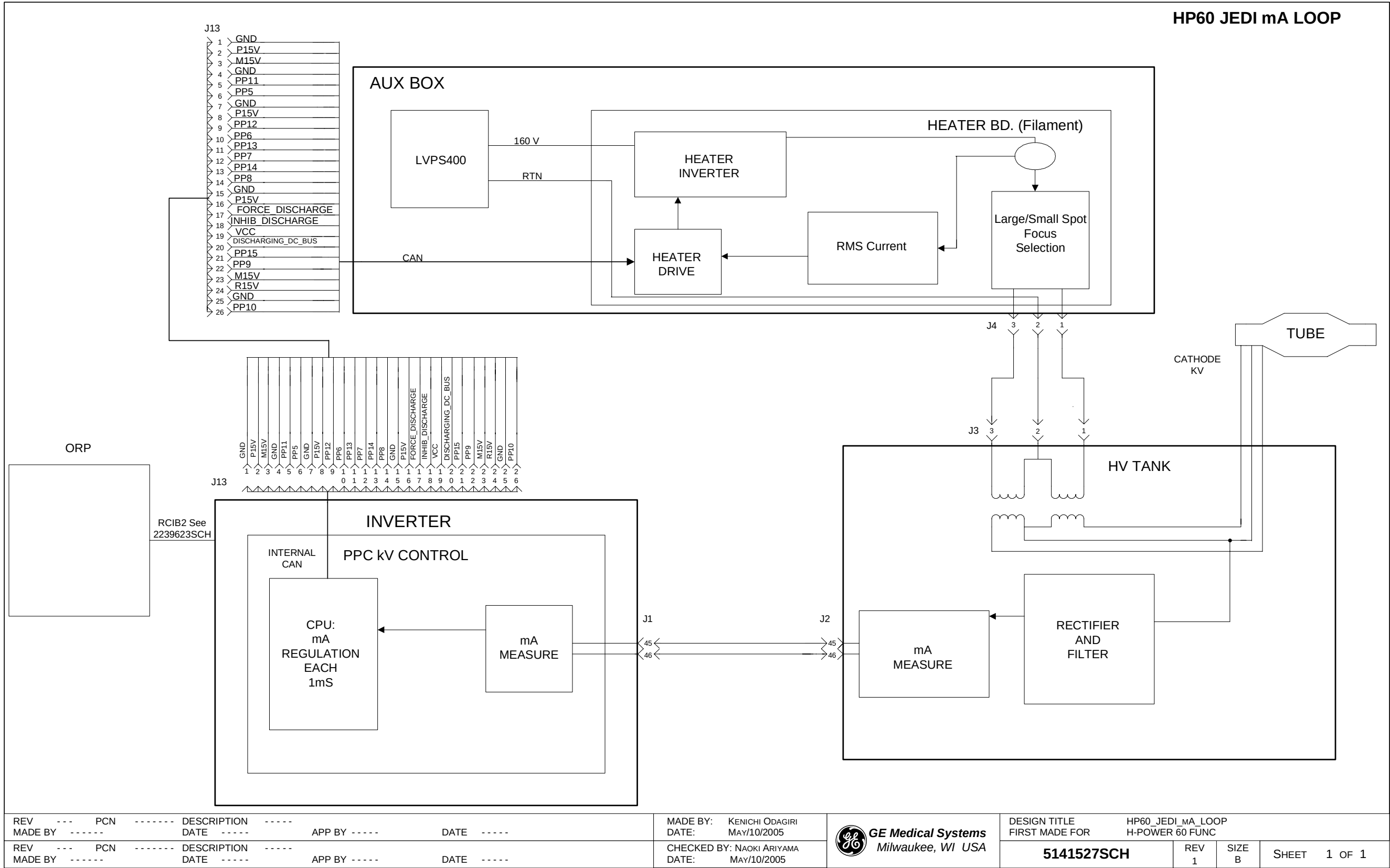
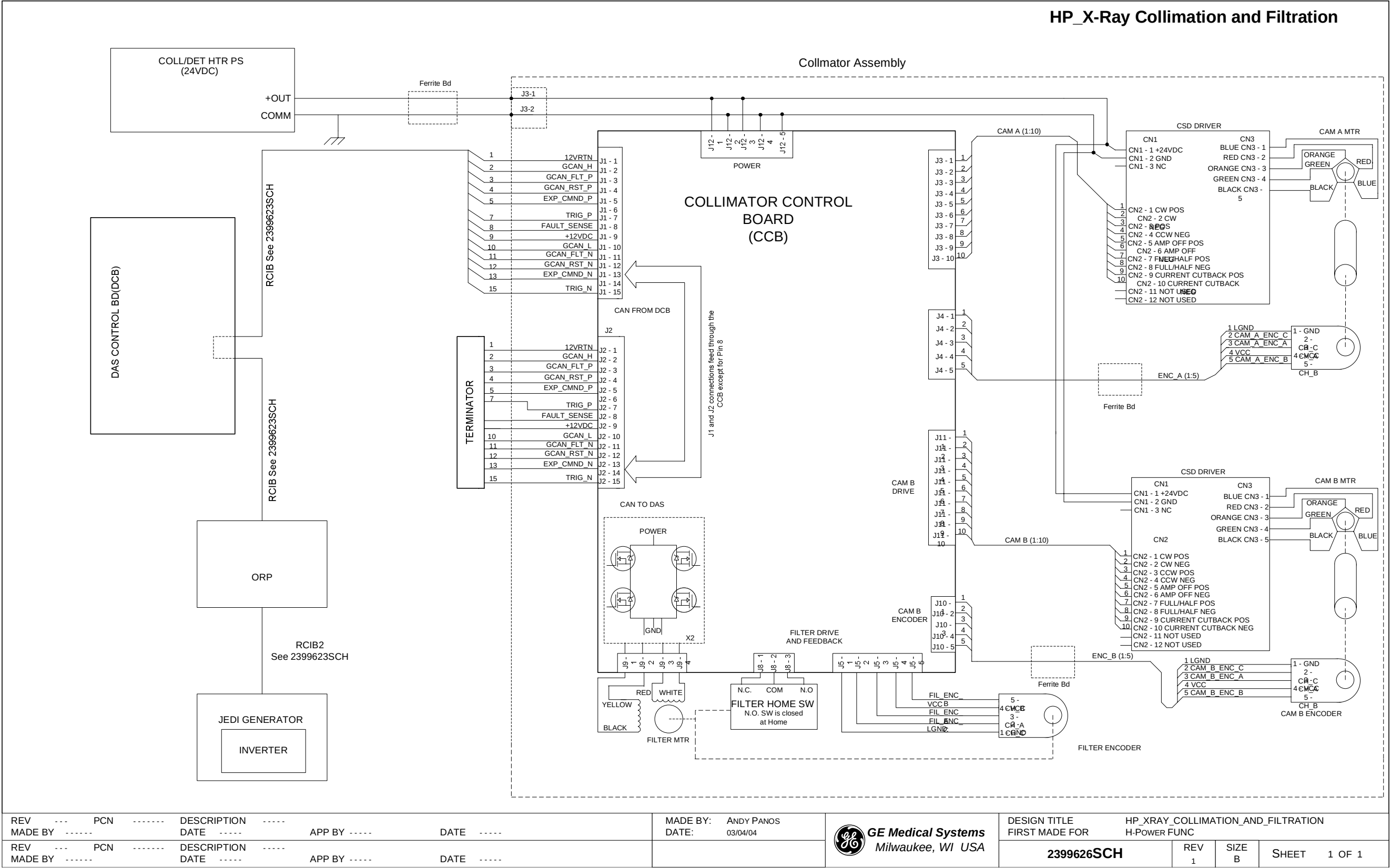


Figure 2-6 5141527 Sheet 1, Rev 1

Section 7.0 — X-Ray Collimation & Filtration - 2399626



Section 8.0 — Exposure Enable and Final Command - 2399629

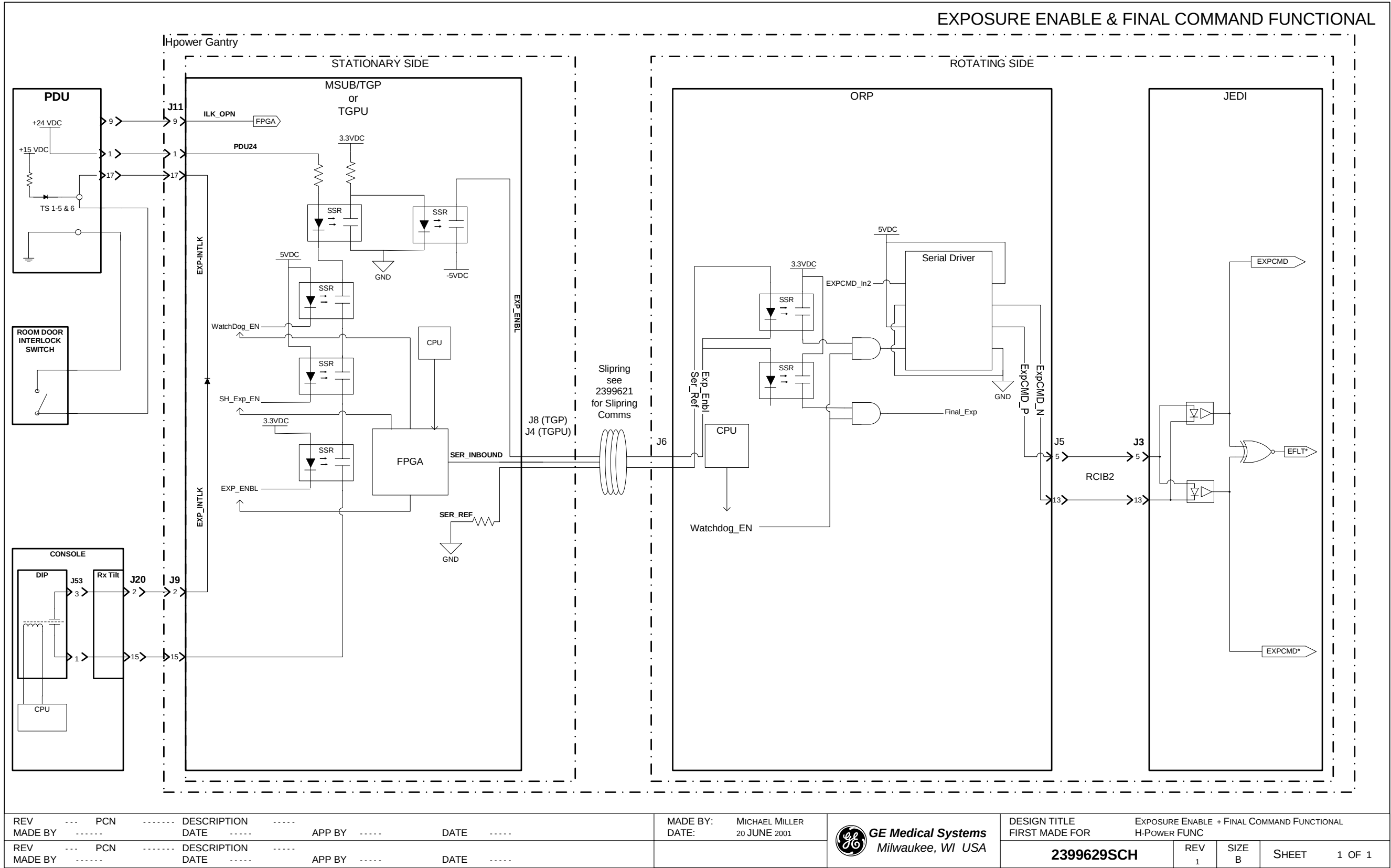


Figure 2-8 2399629 Sheet 1, Rev. 1

REV	---	PCN	-----	DESCRIPTION	-----						MADE BY:	MICHAEL MILLER	 GE Medical Systems Milwaukee, WI USA	DESIGN TITLE		EXPOSURE ENABLE + FINAL COMMAND FUNCTIONAL						
MADE BY	-----			DATE	-----	APP BY	-----	DATE	-----		DATE:	20 JUNE 2001		FIRST MADE FOR		H-POWER FUNC						
REV	---	PCN	-----	DESCRIPTION	-----									 GE Medical Systems Milwaukee, WI USA	2399629SCH		REV	1	SIZE	B	SHEET	1 OF 1
MADE BY	-----			DATE	-----	APP BY	-----	DATE	-----													

Section 9.0 — Tube Rotor Control - 2399630

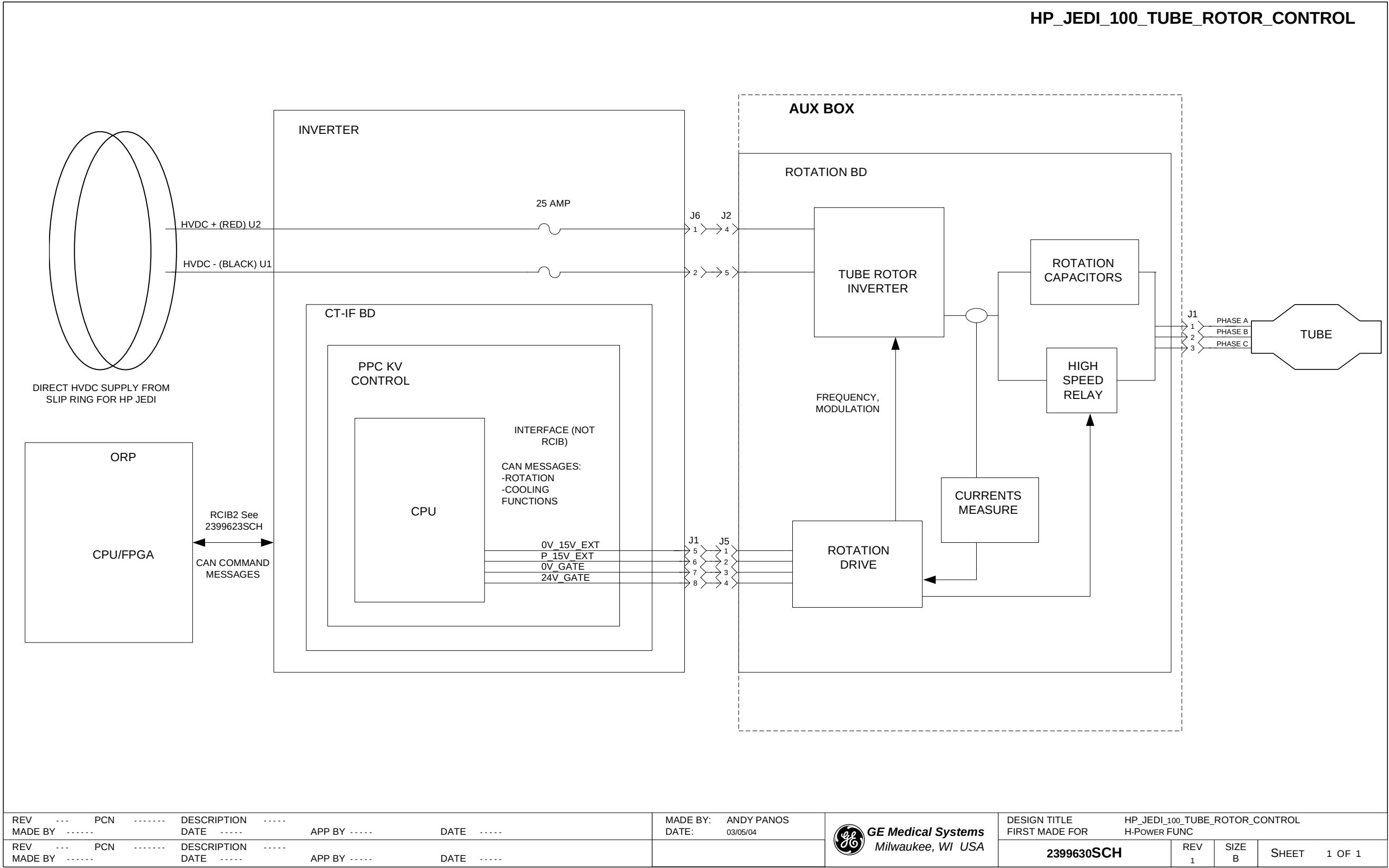
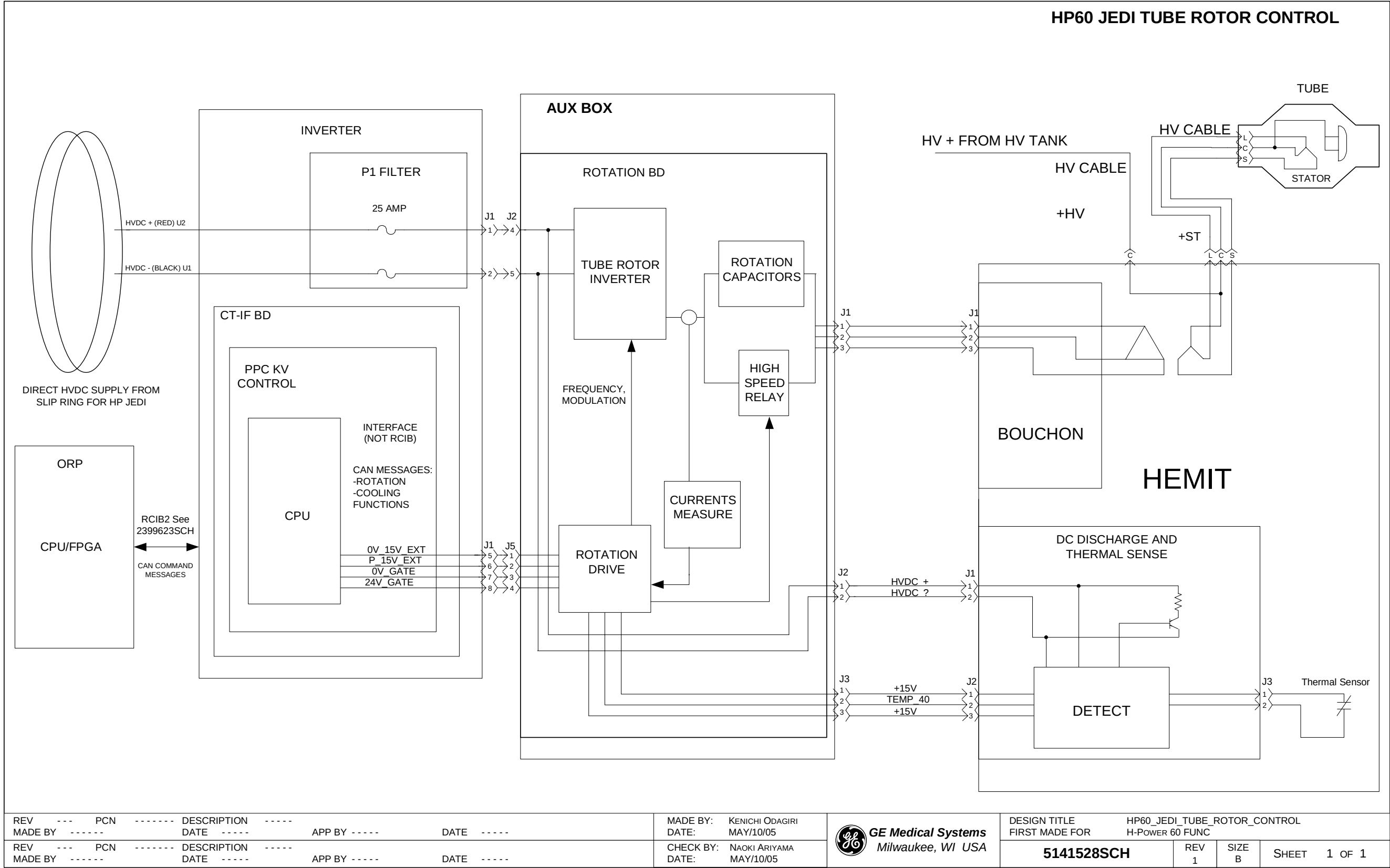


Figure 2-9 2399630 Sheet 1, Rev. 1

Section 10.0 — HP60 JEDI Tube Rotor Control - 5141528



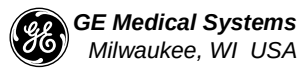
REV --- PCN --- DESCRIPTION --- MADE BY --- DATE --- APP BY --- DATE ---	MADE BY: KENICHI ODAGIRI DATE: MAY/10/05	 GE Medical Systems Milwaukee, WI USA	DESIGN TITLE HP60_JEDI_TUBE_ROTOR_CONTROL FIRST MADE FOR H-POWER 60 FUNC			
REV --- PCN --- DESCRIPTION --- MADE BY --- DATE --- APP BY --- DATE ---	CHECK BY: NAOKI ARIYAMA DATE: MAY/10/05		5141528SCH	REV 1	SIZE B	SHEET 1 OF 1

Figure 2-10 5141528 Sheet 1, Rev 1

Section 11.0 — HP X-Ray Tube Coolant Flow - 2399628

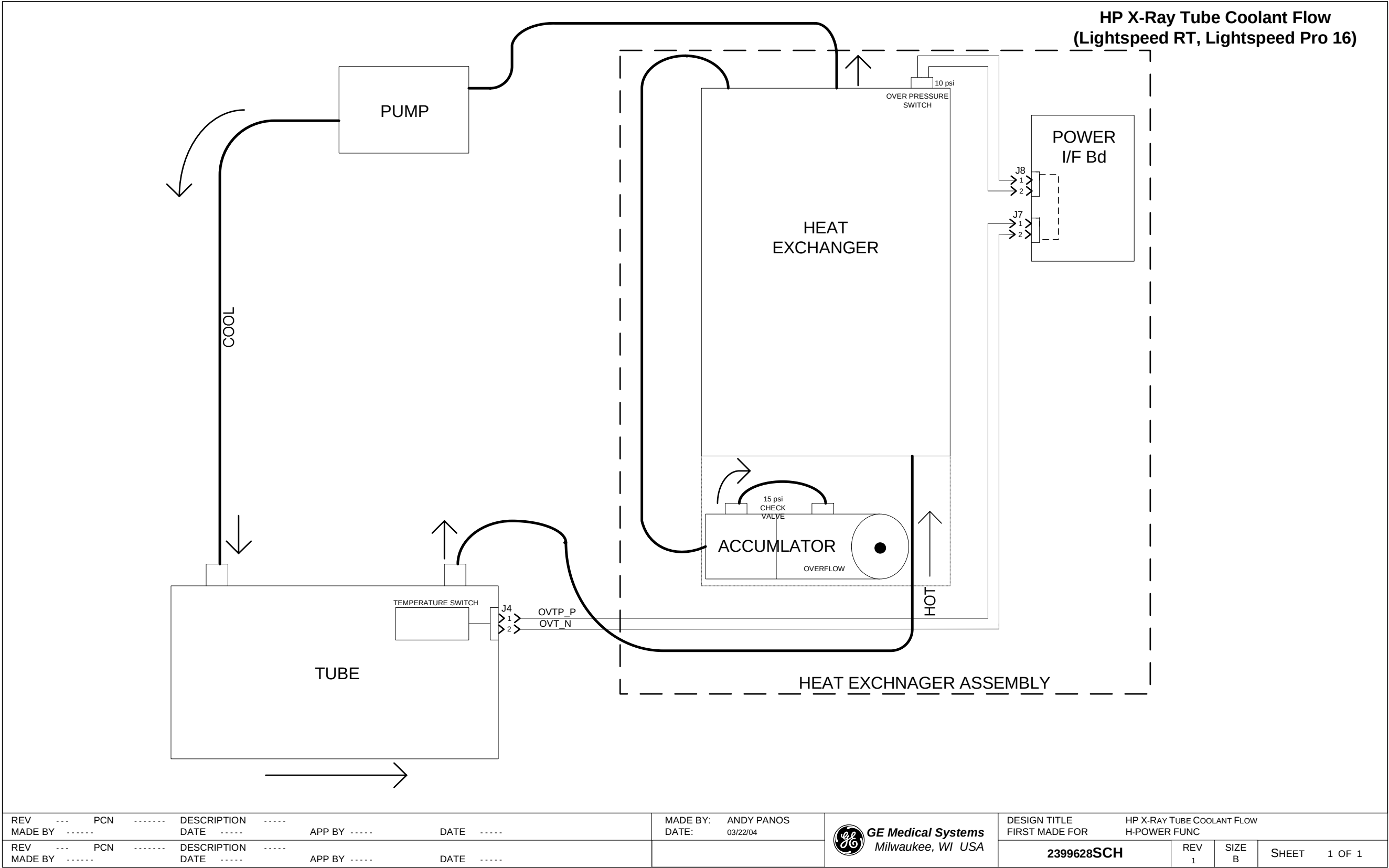
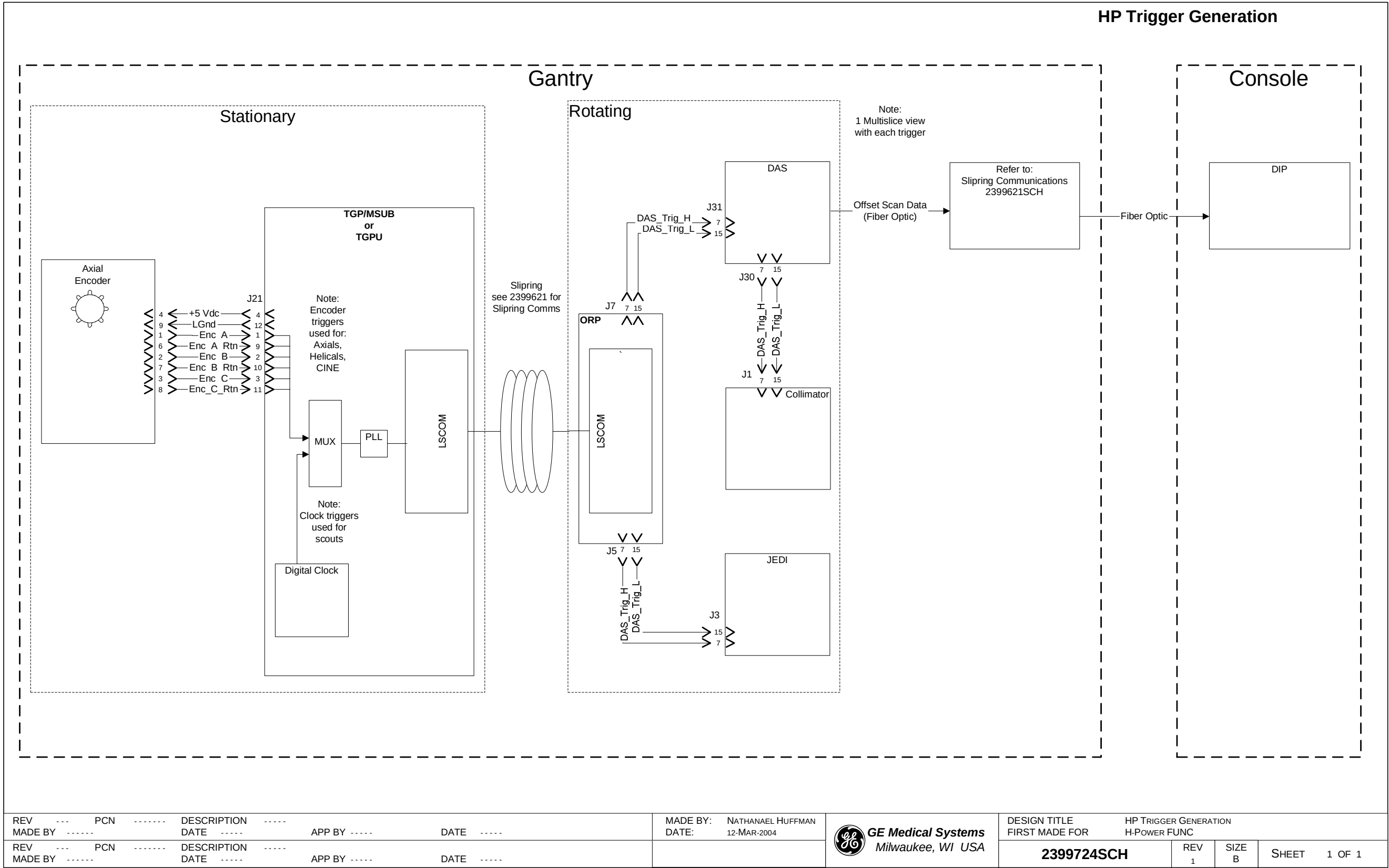


Figure 2-11 2399628 Sheet 1, Rev. 1

Chapter 3

Data Acquisition

Section 1.0 — Trigger Generation - 2399724



REV --- PCN ----- DESCRIPTION -----	MADE BY: NATHANAEL HUFFMAN	DESIGN TITLE HP TRIGGER GENERATION
MADE BY ----- DATE ----- APP BY ----- DATE -----	DATE: 12-MAR-2004	FIRST MADE FOR H-POWER FUNC
REV --- PCN ----- DESCRIPTION -----	GE Medical Systems Milwaukee, WI USA	2399724SCH
MADE BY ----- DATE ----- APP BY ----- DATE -----		REV 1 SIZE B SHEET 1 OF 1

Figure 3-1 2399724 Sheet 1, Rev. 1

Section 2.0 — Detector Heater (DAS Item) - 2399725

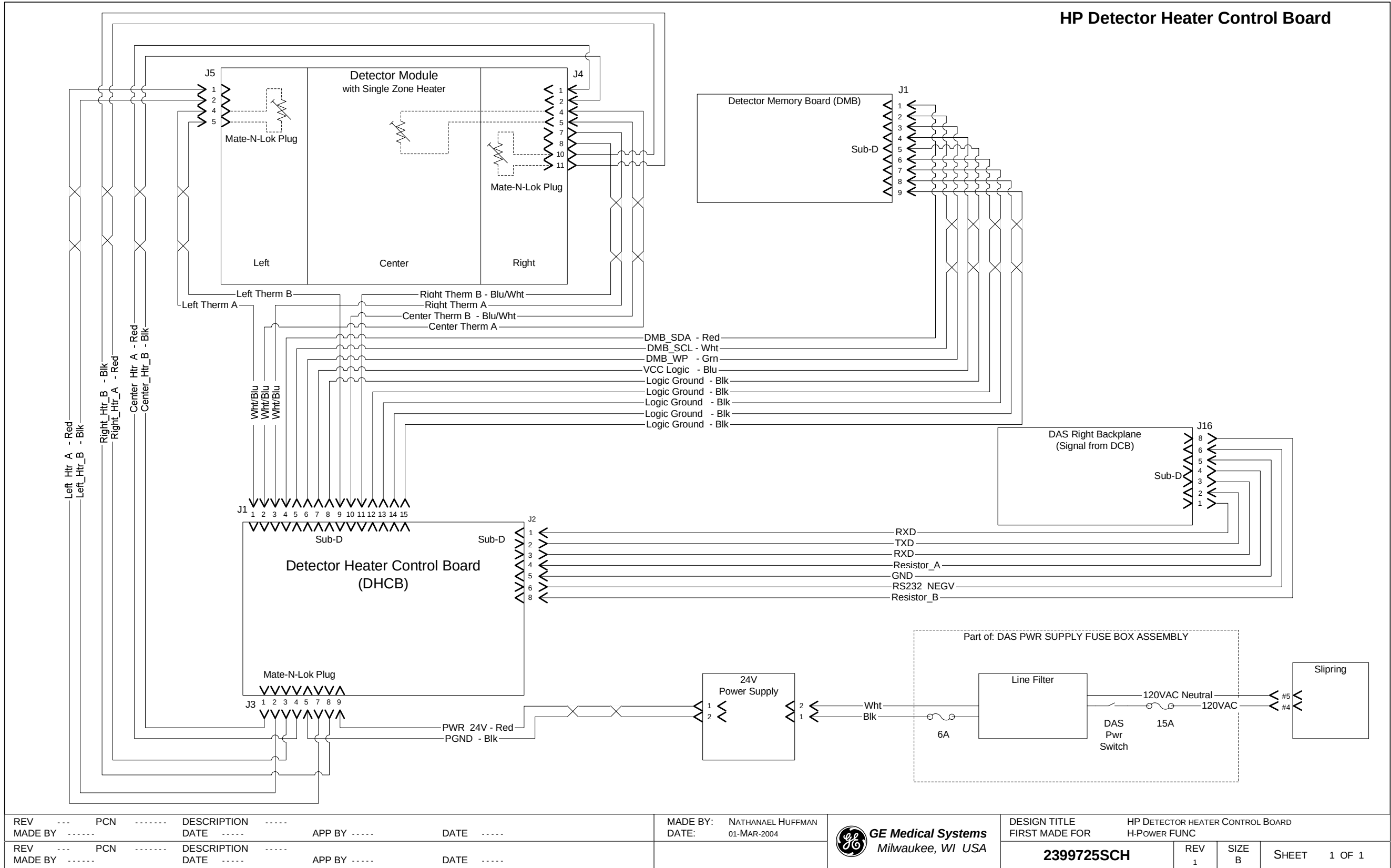


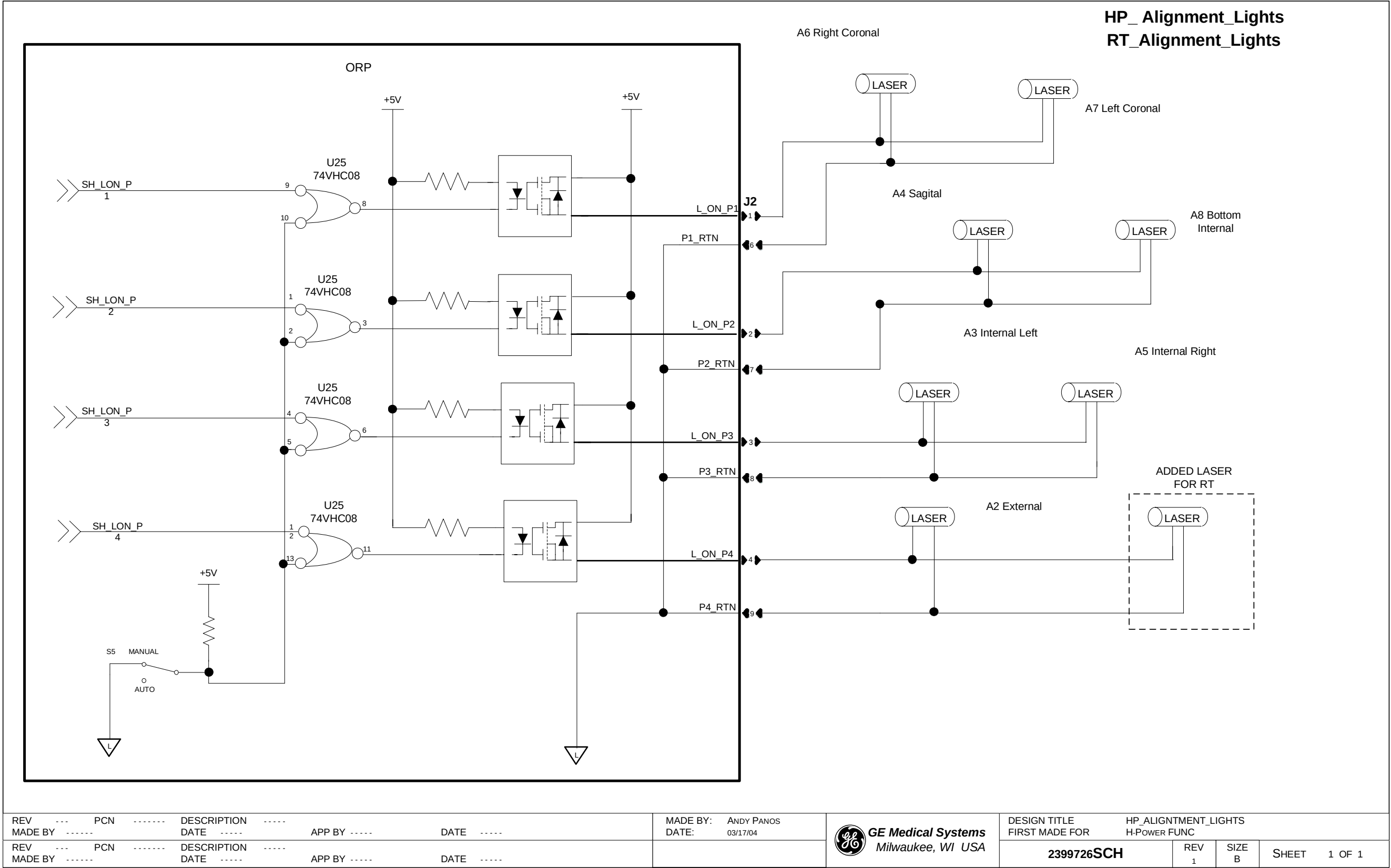
Figure 3-2 2399725 Sheet 1, Rev. 1

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Chapter 4

Patient Positioning

Section 1.0 — Alignment Lights - 2399726



Section 2.0 — Gantry Tilt, from Gantry, Console - 2399727

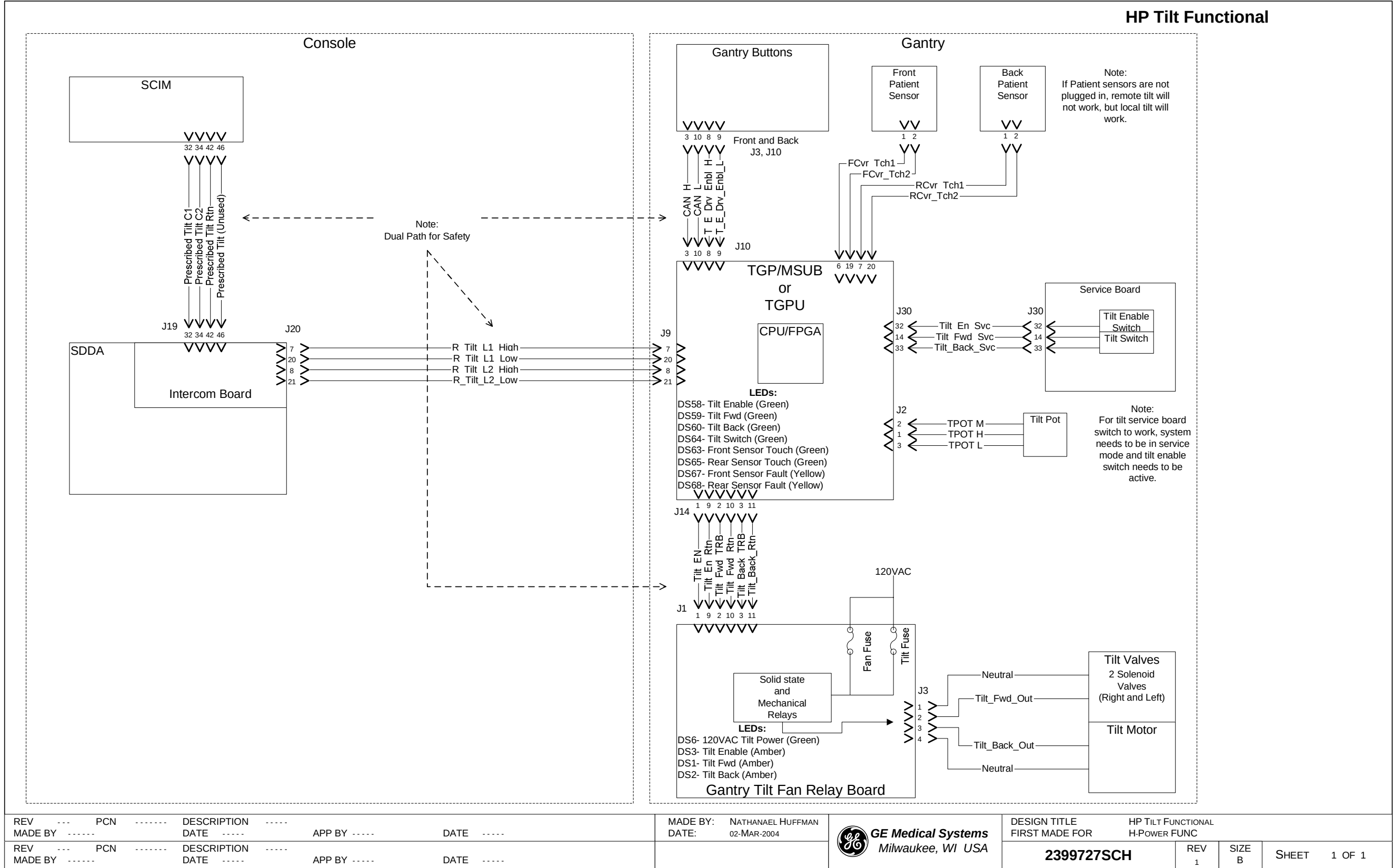


Figure 4-2 2399727 Sheet 1, Rev. 1

Section 3.0 — Table Cradle Sync - 2399728

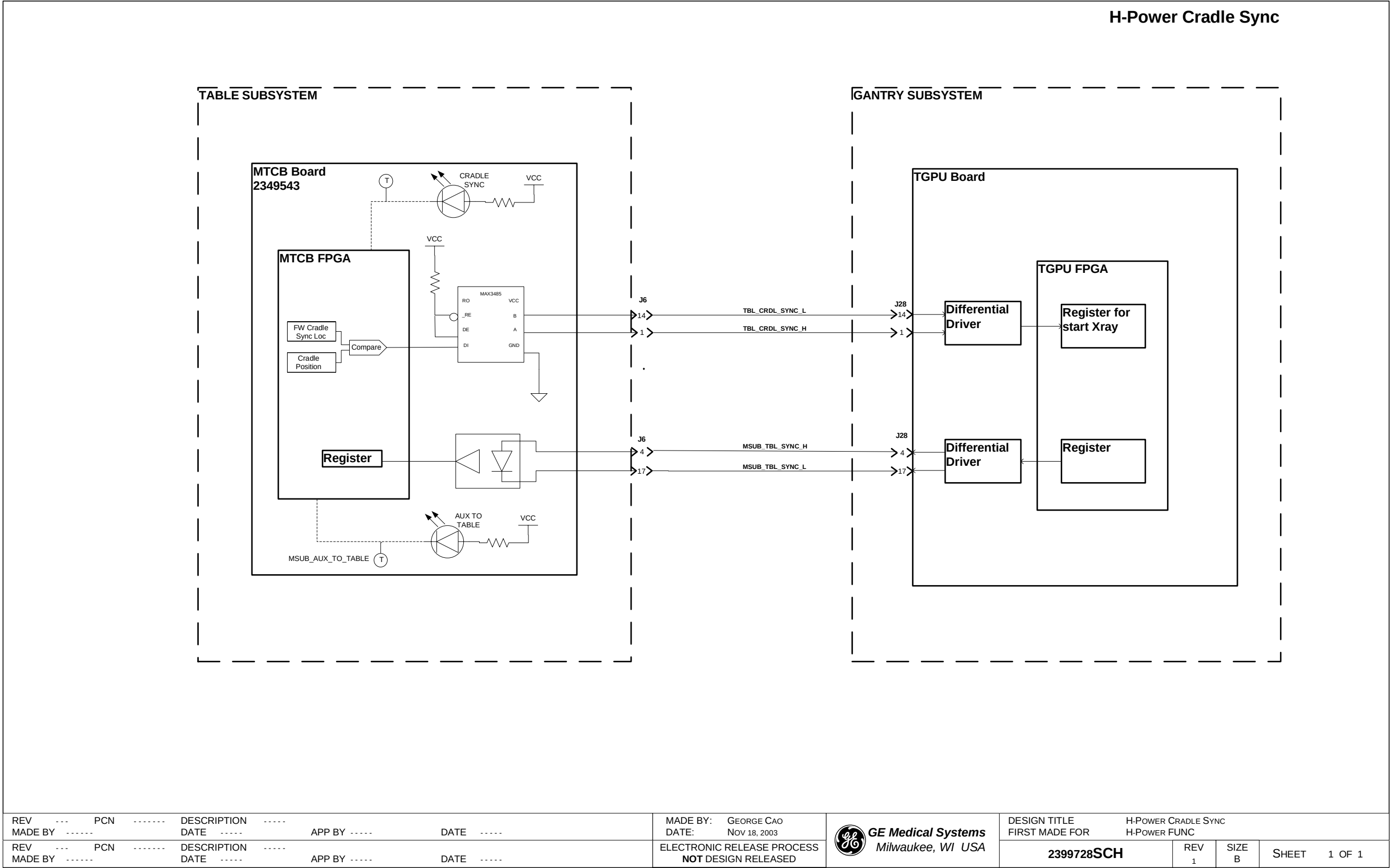
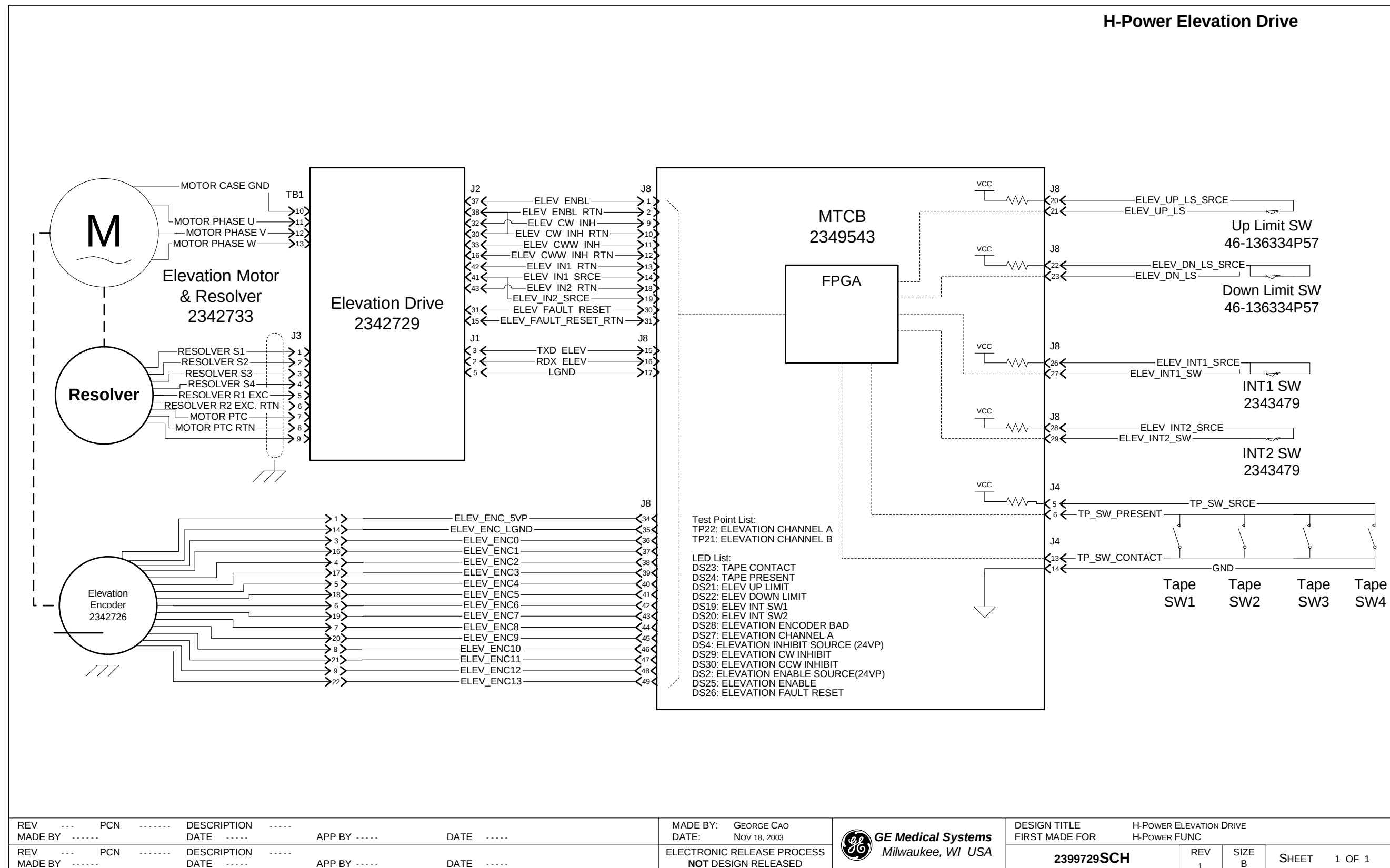


Figure 4-3 2399728 Sheet 1, Rev. 1

Section 4.0 — Table Elevation - 2399729



Section 5.0 — Table Cradle Drive - 2399730

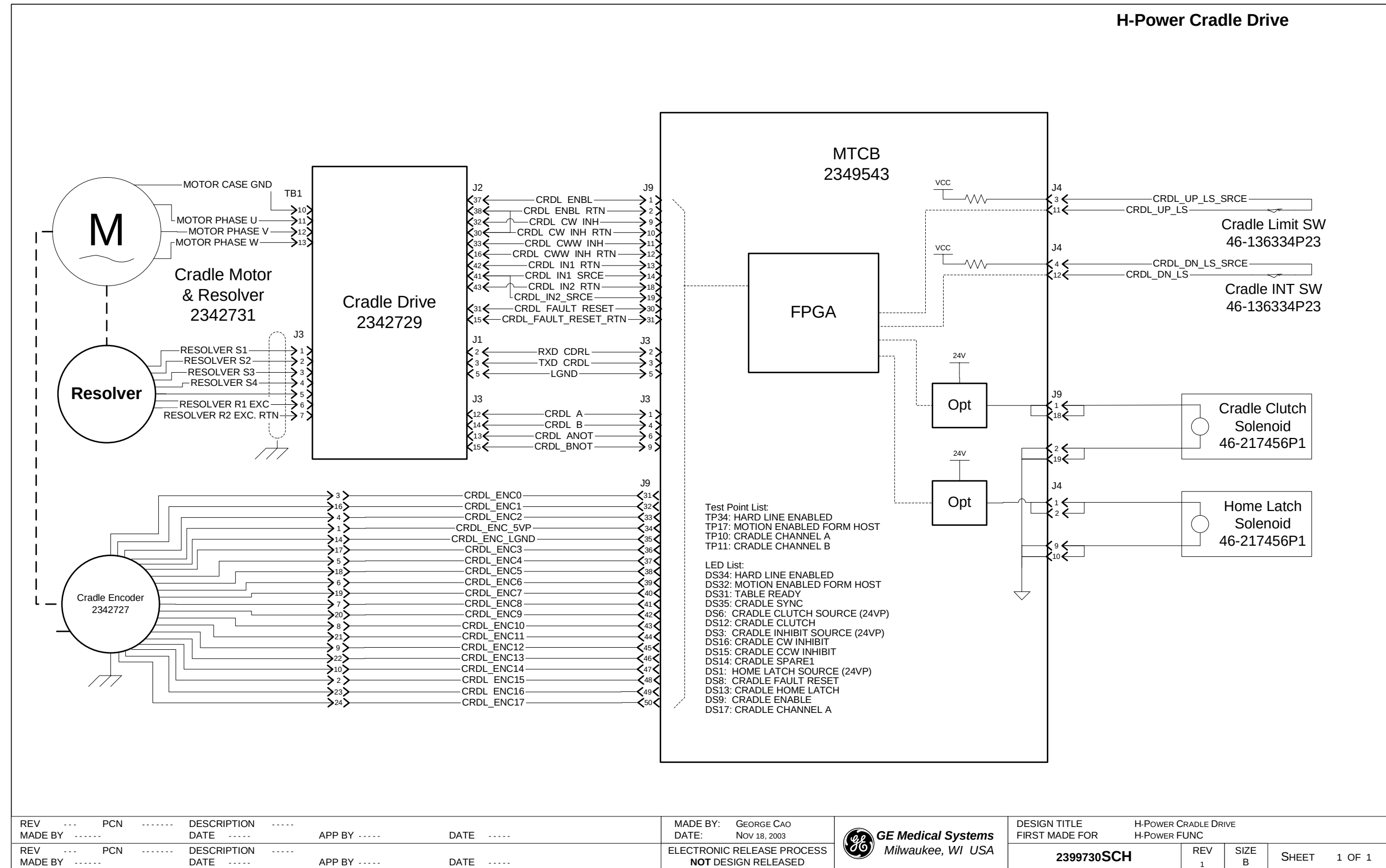


Figure 4-5 2399730 Sheet 1, Rev. 1

Chapter 5

Axial Control

Section 1.0 — Axial Servo Control / Dynamic Brake - 2399731

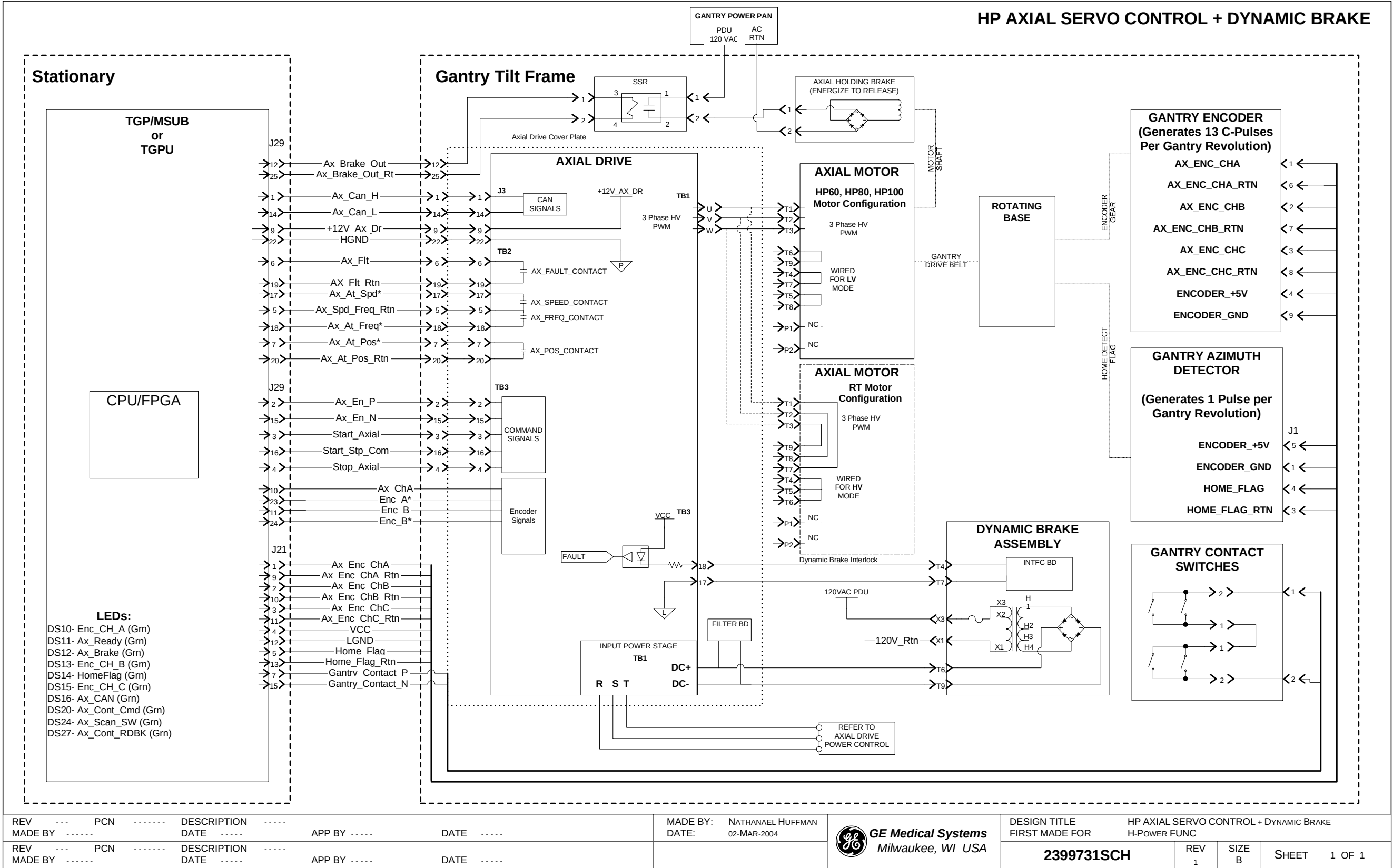


Figure 5-1 2399731 Sheet 1 Rev. 1

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Chapter 6

Operator I/O

Section 1.0 — Gantry Operator Interface and Displays - 2399733

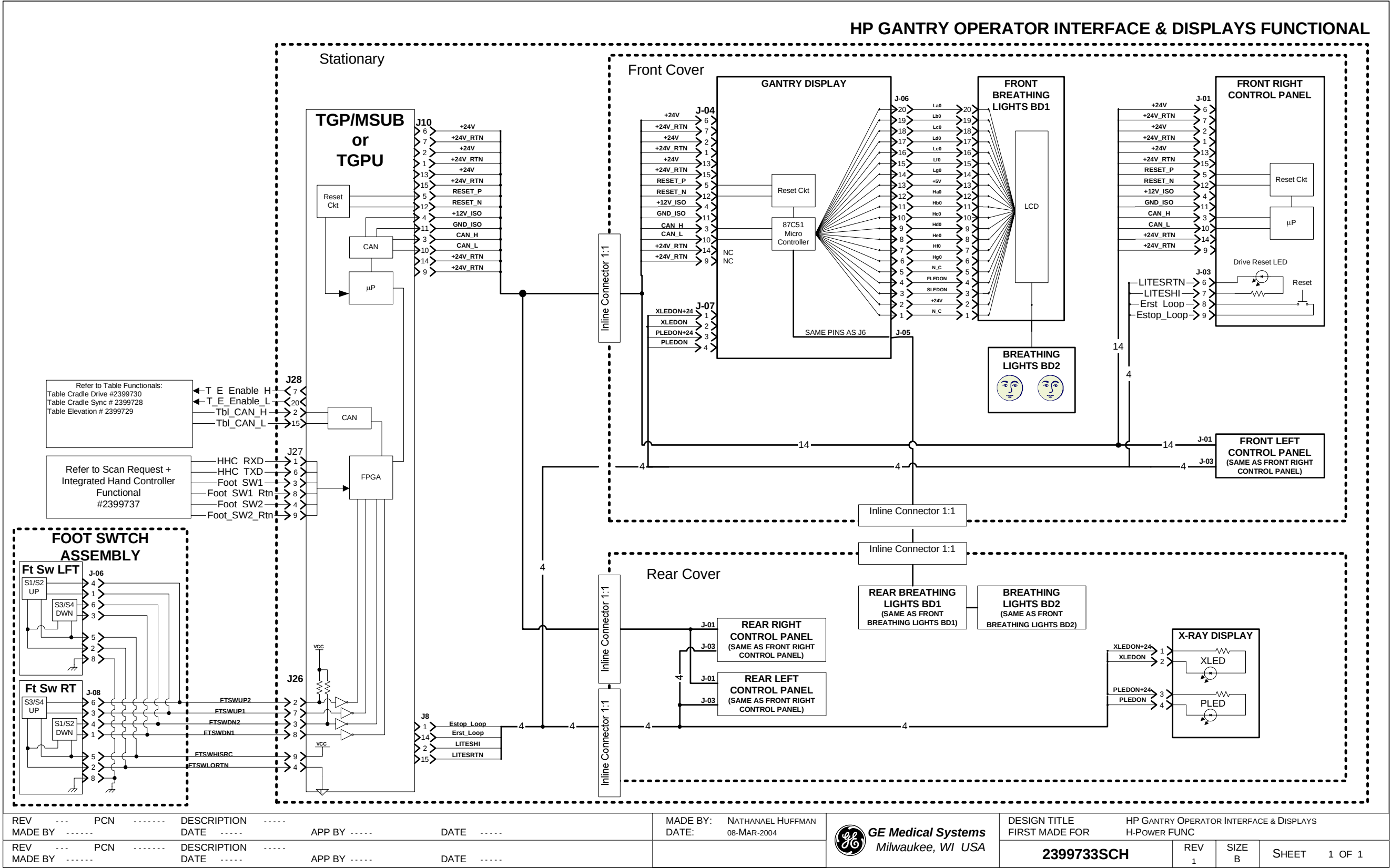


Figure 6-1 2399733 Sheet 1, Rev. 1

Section 2.0 — Gantry Reset from Console - 2399734

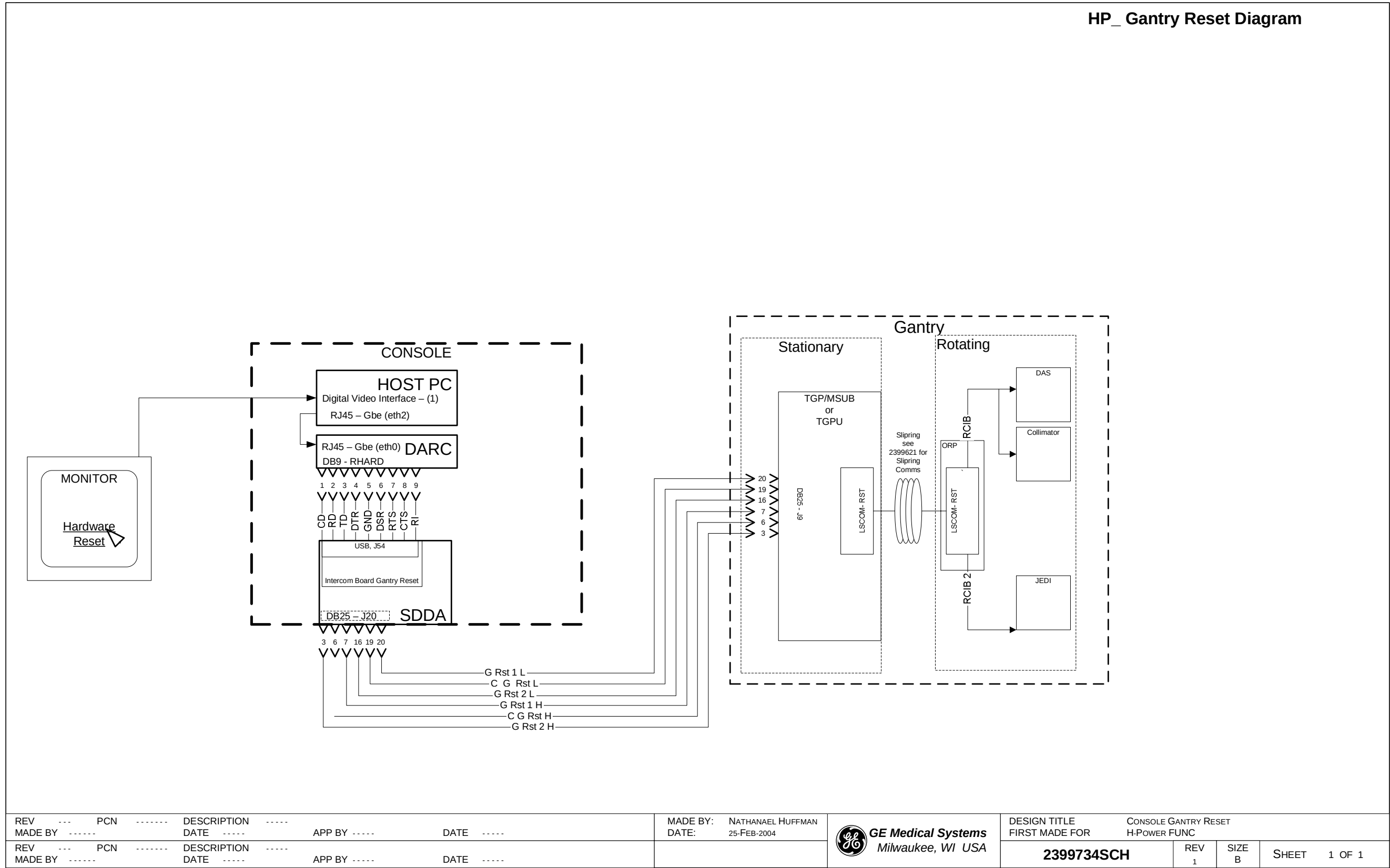
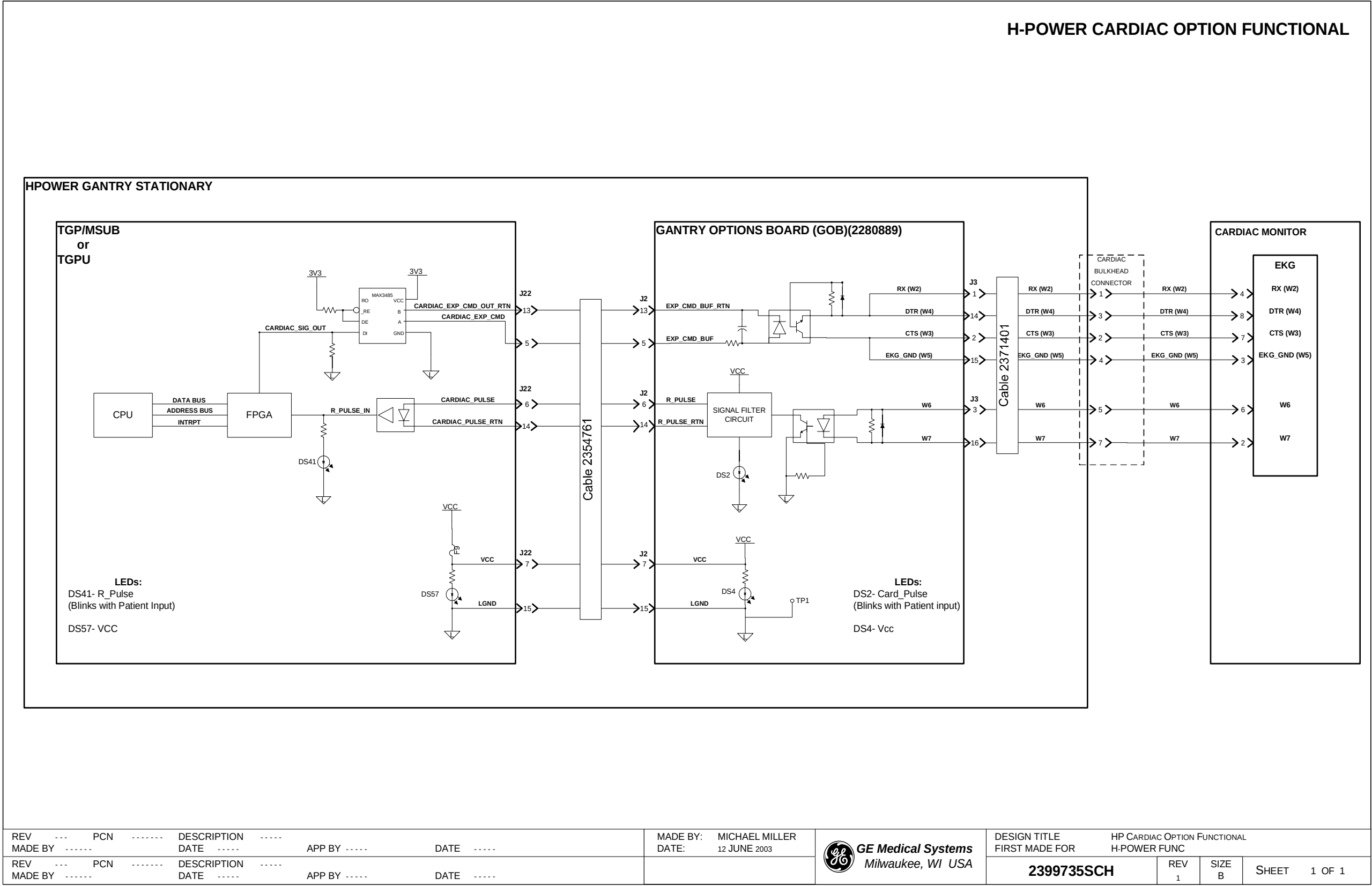


Figure 6-2 2399734 Sheet 1, Rev. 1

Section 3.0 — Cardiac Options - 2399735



Section 4.0 — Pulmonary Options - 2399736

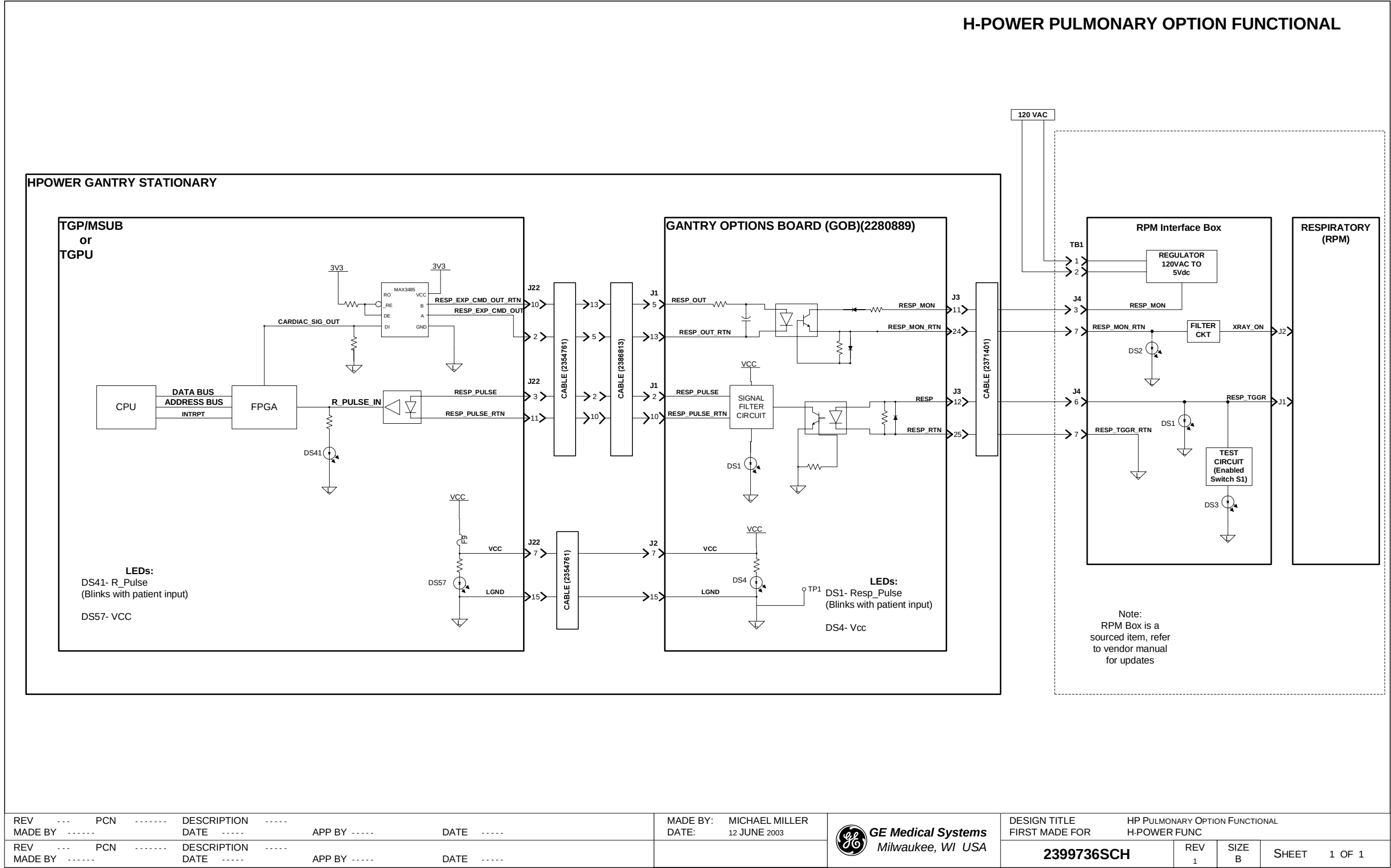


Figure 6-4 2399736 Sheet 1, Rev. 1

Section 5.0 — Scan Footswitch and IHHC Controller - 2399737

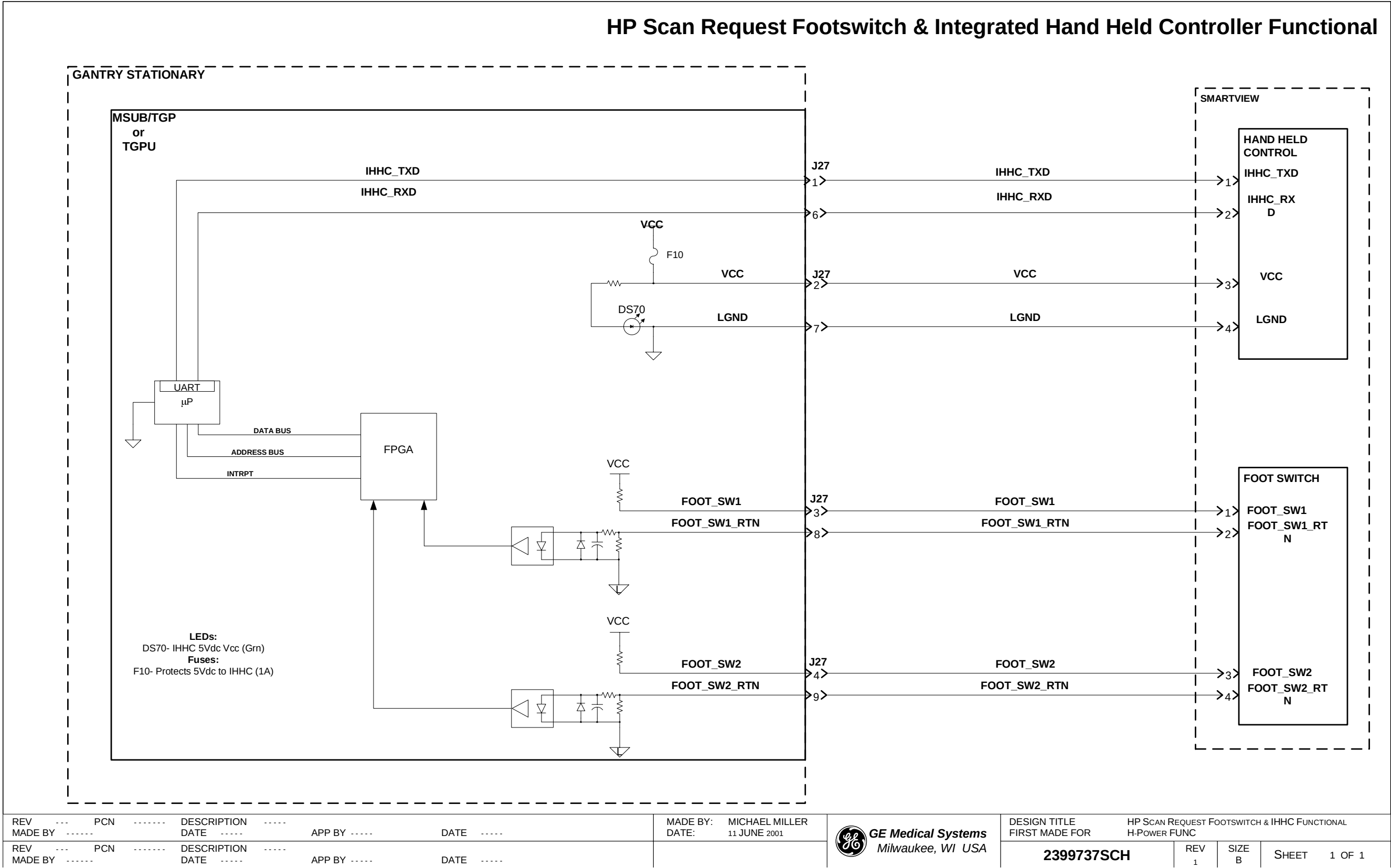


Figure 6-5 2399737 Sheet 1, Rev. 1

Section 6.0 — Site X-Ray Light, System Light - 2399738

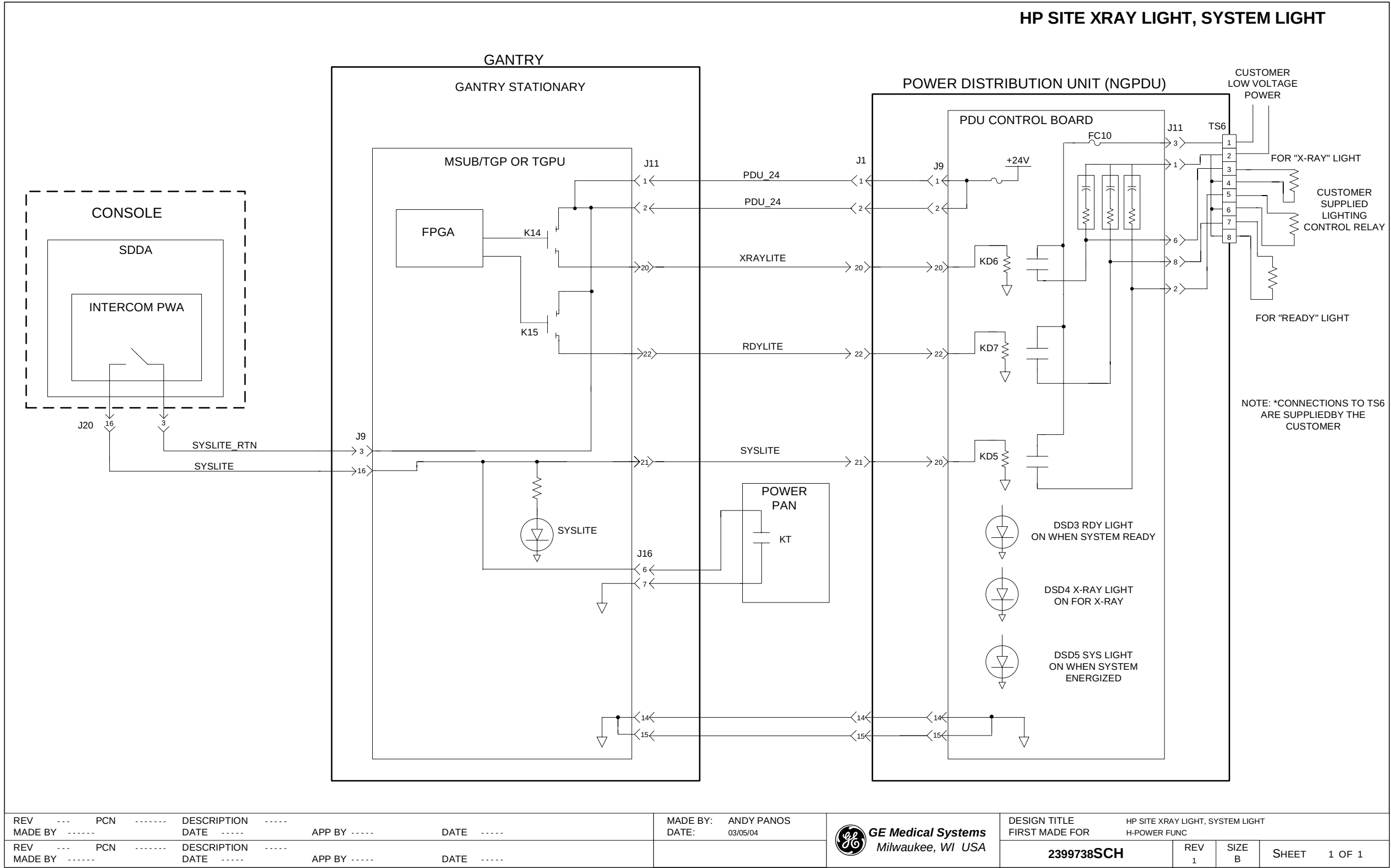


Figure 6-6 2399738 Sheet 1, Rev. 1

Section 7.0 — Autovoice / Intercom - 2399739

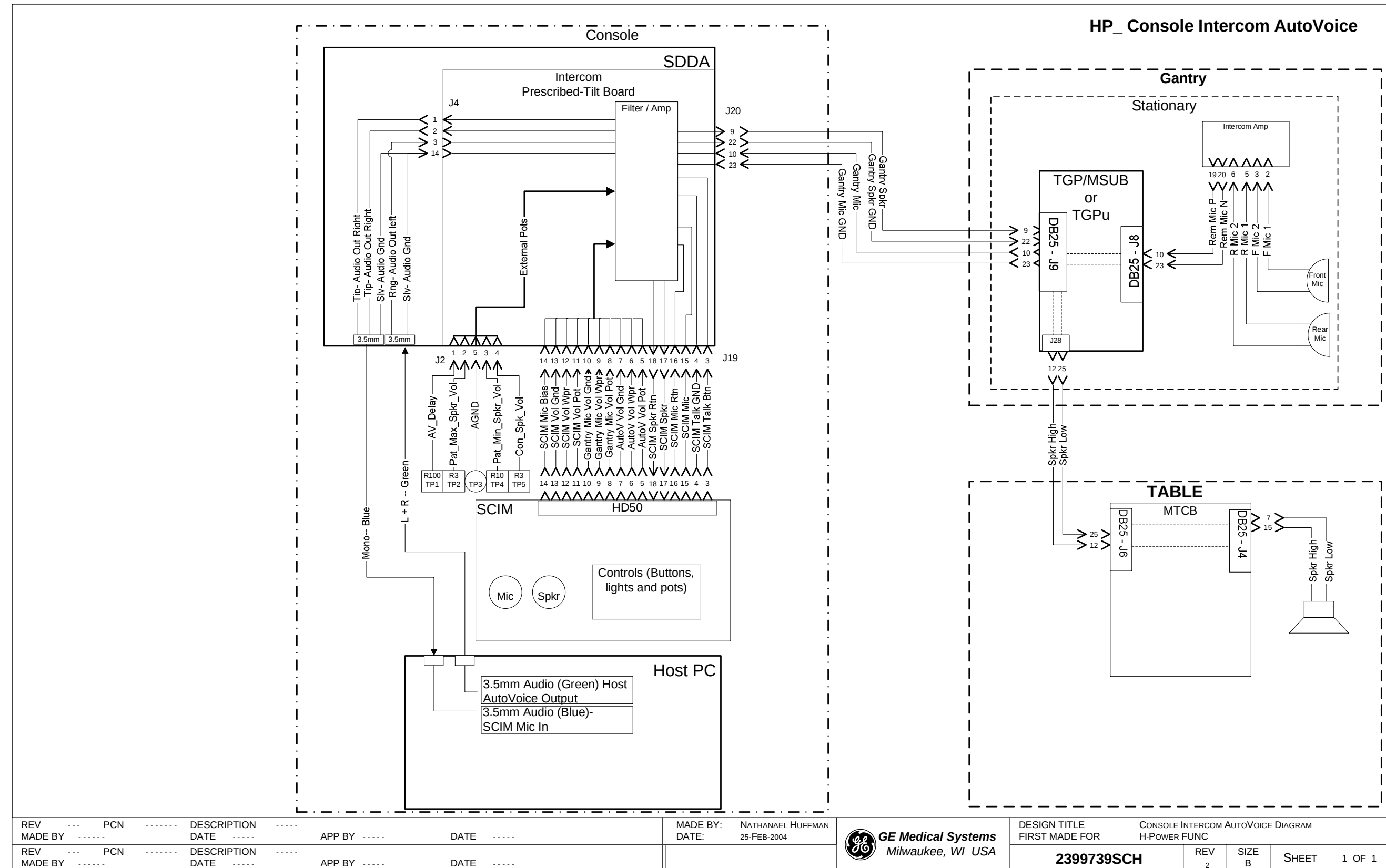


Figure 6-7 2399739 Sheet 1, Rev. 2

Chapter 7

System Monitoring

Section 1.0 — Gantry Temperature and Balance Sensors - 2399740

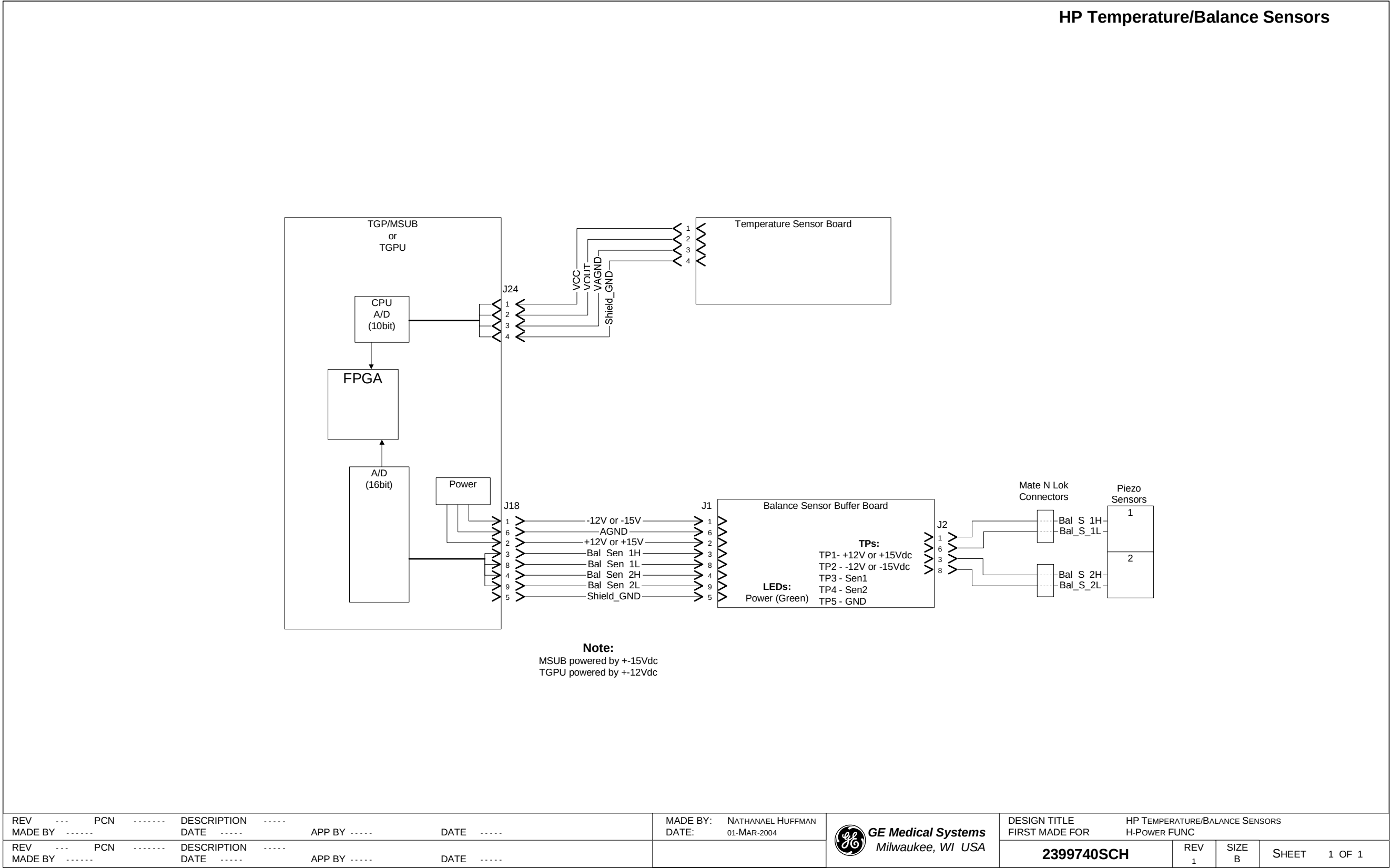


Figure 7-1 2399740 Sheet 1, Rev. 1

Section 2.0 — Gantry Cover Fans - 2399741

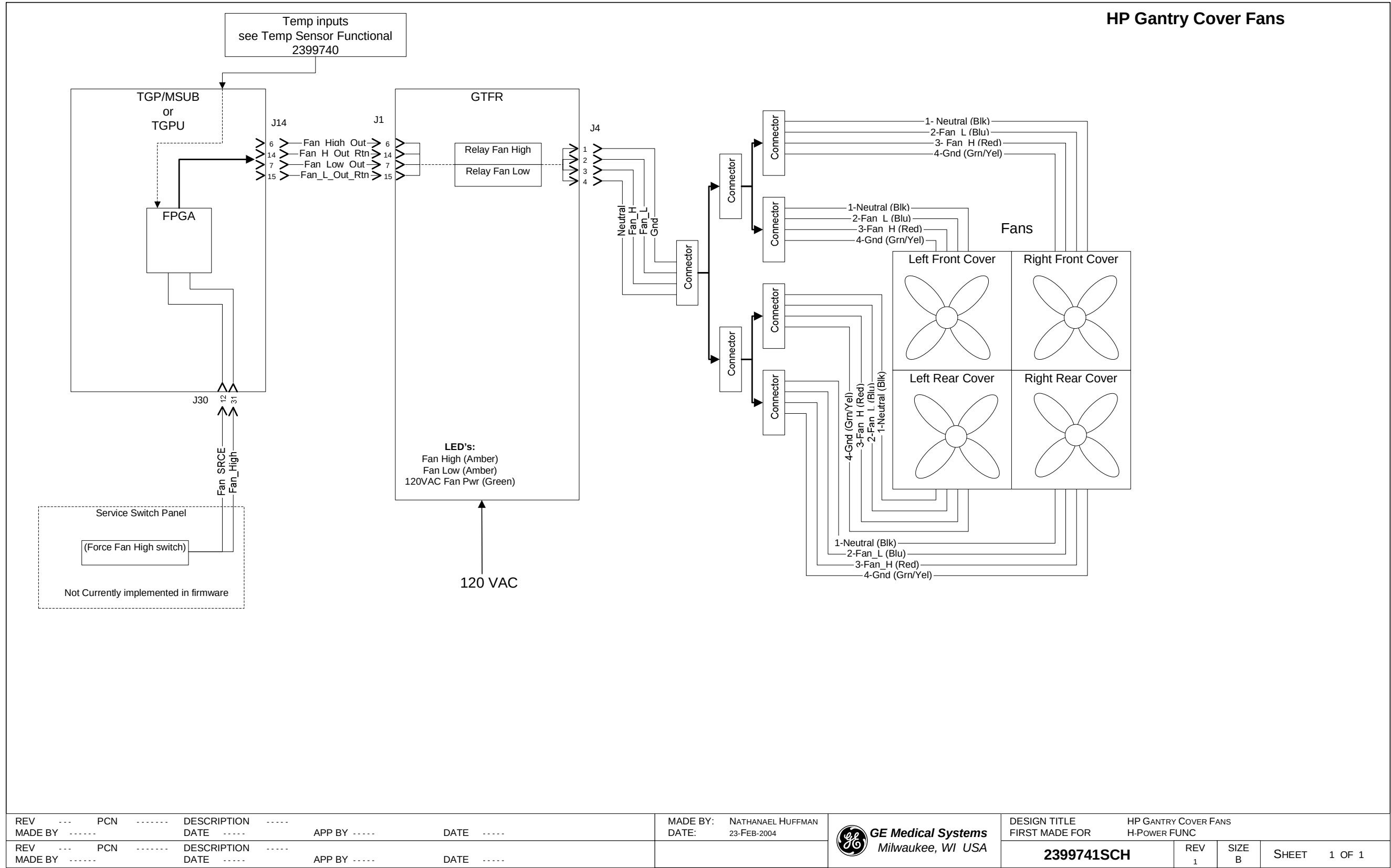


Figure 7-2 2399741 Sheet 1, Rev. 1

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Chapter 8

System Power Control

Section 1.0 — System E-Stop and Power Interlocks - 2399742

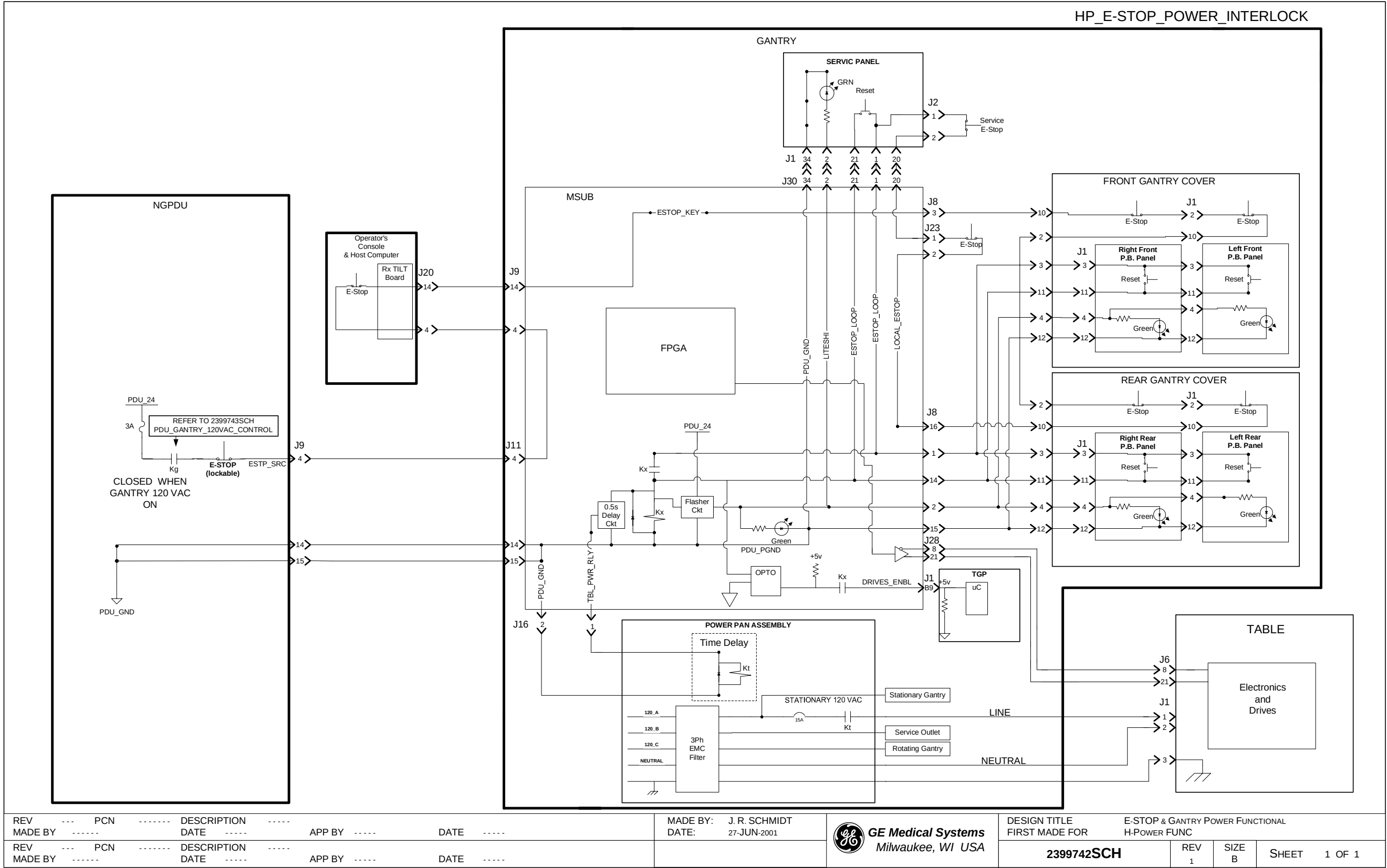


Figure 8-1 2399742 Sheet 1, Rev. 1

Section 2.0 — PDU - Gantry 120 VAC Control - 2399743

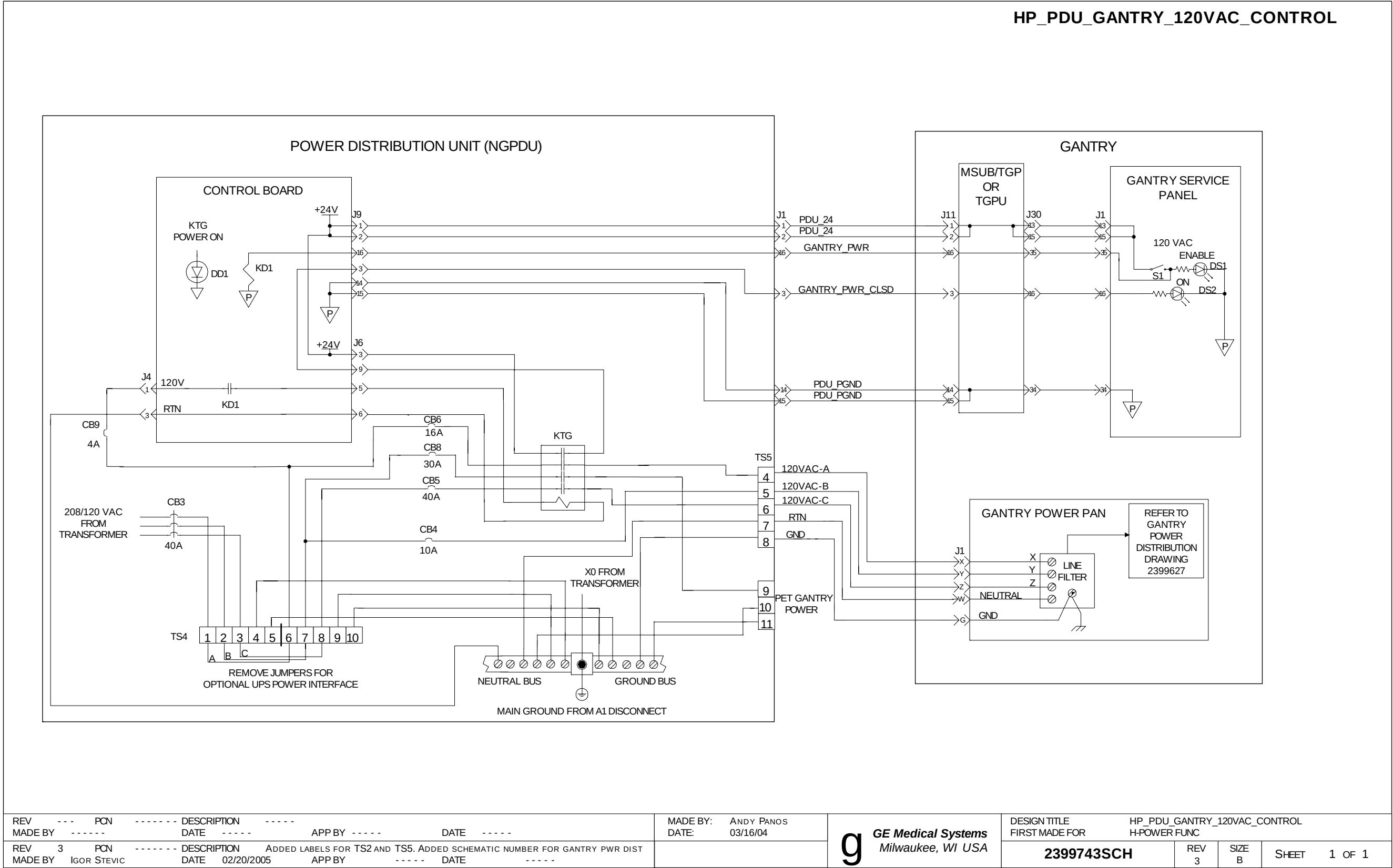
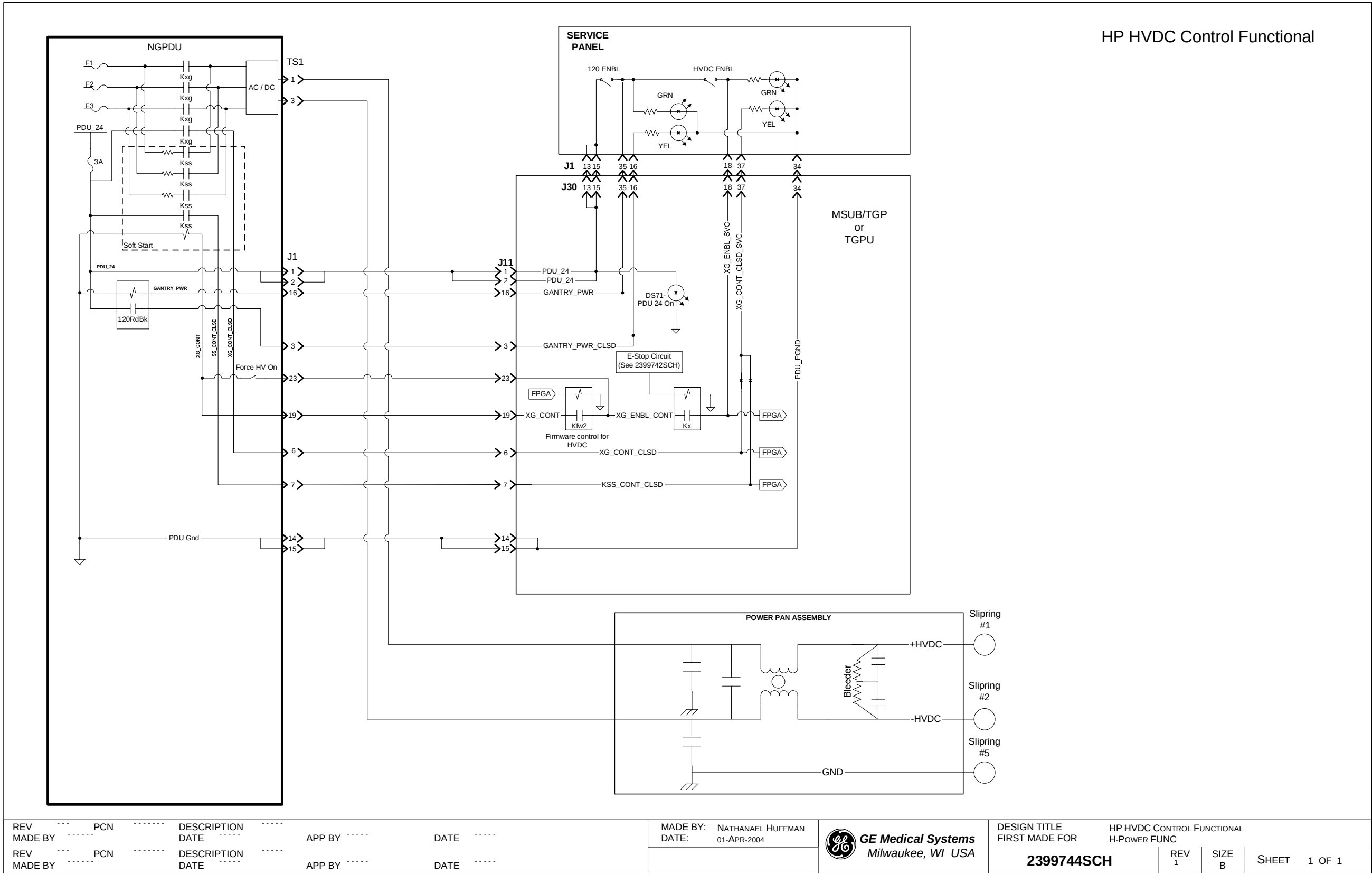


Figure 8-2 2399743 Sheet 1, Rev. 3

Section 3.0 — HV_DC_Control - 2399744



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